Balancing College and Work Activities: The Gig Economy Meets Online Education*

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Abstract

We study how students balance college and work when both offer flexible schedules, using high-frequency data from an Arizona State University-Uber partnership that lets drivers take free online courses. An additional hour of coursework completed by classmates–a proxy for course demands–displaces about two-thirds of an hour on the Uber platform (1.3 fewer trips). Forgone earnings are modest ($41 per week, under 10% of weekly earnings), while study time barely responds to short-run labor-market conditions. Students re-time work as course demands rise, with no detectable reduction in course grades from added work hours. Survey respondents value flexible formats and expect meaningful returns.

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1 Introduction

Student employment is a central feature of higher education in the United States. In 2019, 45% of full-time and 84% of part-time undergraduate students worked for pay (US Department of Education, 2020). Among working students, employment was also intensive: 64% of full-time and 86% of part-time undergraduates worked at least 20 hours per week. These patterns are not new. The average labor supply of 18- to 22-year-old full-time undergraduates nearly doubled between 1970 and 2000 (Scott-Clayton, 2012), and high levels of student employment are likely to remain an important feature of college attendance. Two broader forces point in this direction. First, the cost of college has risen substantially, increasing the need for many students to finance their education through work.[1] Second, technological change has altered labor-market demands, reducing opportunities in low-skill and routine jobs and encouraging workers to acquire new skills while remaining employed (Saliola et al., 2020). The rise of artificial intelligence may reinforce this pressure. Given that AI is expected to affect nearly 40% of jobs worldwide (Cazzaniga et al., 2024), even many high-skill workers may need to update their skills in response to changing skill demands.

At the same time, existing evidence suggests that working while attending college meaningfully crowds out schooling time, which often negatively affects college performance (Stinebrickner and Stinebrickner, 2003; Beffy et al., 2010; Kalenkoski and Pabilonia, 2010; Scott-Clayton, 2011, 2012; Darolia, 2014; Scott-Clayton and Minaya, 2016; Soliz and Long, 2016; Neyt et al., 2019; Le Barbanchon et al., 2023).[2] Taken together, these facts highlight the potential value of new alternatives that could help students balance both activities.

The aim of this study is twofold. First, we examine how students allocate time between coursework and paid work in a setting where both activities offer substantial scheduling flexibility. Specifically, we examine whether short-run, within-semester work-study tradeoffs remain economically meaningful by estimating how labor supply responds to course demands and how study effort responds to earnings opportunities. A large literature studies how student work affects academic performance; however, little is known about how college

[1] Between 2006 and 2020, after adjusting for inflation, the average cost of attendance at public and non-profit private four-year universities increased by almost 20% (Ma and Pender, 2022).

[2] These studies typically measure the cumulative or annual displacement of schooling by work, rather than short-run, within-semester reallocation effects that we focus on.

demands affect labor supply, or how short-run earnings opportunities affect study effort. Thus, our primary contribution is to provide the first high-frequency estimates of these trade-offs, exploiting weekly within-individual variation in coursework and local labor-market wages.

Second, we turn from short-run time allocation to the broader value students attach to this highly flexible model of work and schooling. Using survey evidence, we examine students' expected returns to completing an online bachelor's degree, their stated likelihood of enrolling under alternative combinations of learning modality, work-schedule flexibility, and tuition cost, as well as their prior attachment to higher education. Using administrative records on credit accumulation and graduation, we also provide evidence on students' realized progress toward degree completion.

We study these two margins in the context of a partnership between Arizona State University (ASU) and Uber, which allows eligible drivers to enroll in ASU Online classes tuition-free; henceforth, we refer to drivers enrolled in the ASU program as Uber-students.[3] The partnership combines the two forms of flexibility that are central to our question: students can complete online coursework with substantial scheduling flexibility while also adjusting their Uber labor supply at high frequency.

Although this is a specific institutional setting, the underlying combination of flexible work and flexible online learning lies at the intersection of broader changes in both education and work. The share of postsecondary students taking distance courses rose from 28% in 2014 to 37% in 2019 (U.S. Department of Education, National Center for Education Statistics, 2020), while the gig economy workforce expanded by more than 300% between 2015 and 2021 (Denes et al., 2025). Together, these trends suggest that settings in which students combine flexible work and flexible online learning may become increasingly important in higher education. Thus, the ASU–Uber partnership is not only useful for measurement and identification; it also provides evidence on a work-study arrangement that may become increasingly common.

Estimating these trade-offs is empirically challenging because work hours and study ef-

[3]Programs in which companies fund their employees' education have become increasingly common in recent years. For instance, Starbucks, Walmart, and Disney also cover the tuition costs for employees pursuing post-secondary education.

fort are rarely measured together at high frequency, even though the relevant trade-offs often occur within weeks or semesters. Moreover, both activities are chosen jointly, making it difficult to separate responses to coursework demands from responses to labor market opportunities. Three key features of our institutional setting allow us to address these challenges. First, Uber-students face a flexible environment where work and study hours can be adjusted frequently. This makes the setting well-suited for studying whether short-run opportunity costs remain meaningful when students have more flexibility to re-time work and study effort. Second, the Uber and ASU Online platforms provide granular, high-frequency information about drivers and students over time, offering a unique opportunity to examine how effort responds to changes in work and college conditions. Third, the degree of complementarity between Uber work and skills acquired in college is likely to be much lower than in other work-study contexts. In other words, college courses are unlikely to improve a student's capabilities as an Uber driver, making the interpretation of our findings cleaner.

For the main high-frequency analysis, we link records from the Uber and ASU Online platforms. The Uber records cover drivers enrolled in fully online bachelor's degree-seeking programs at ASU, as well as a subset of their Uber-driving peers who qualified for but did not enroll in the ASU program. These records include weekly information on completed trips, total pay, minutes on the application, driving habits, city-level labor market conditions, and background characteristics. The ASU records cover the universe of Uber-students and their online classmates who are not program participants. They include student background characteristics, transcript data, and weekly activity measures from the online learning platform, including points earned, assignments submitted, Canvas logins, clicks, and minutes online. Together, these linked records allow us to measure weekly work and study effort for the same individuals at high frequency over time.

Our econometric specifications are derived from a simple analytical framework that justifies our empirical models and helps characterize the costs of performing learning and working activities simultaneously. To address selection and endogeneity concerns, our empirical strategy combines within-individual variation, classroom-level variation in course demands, and instrumental variables based on local labor-market conditions to identify the short-run trade-offs associated with balancing work and learning activities. More specifically, we ex-

ploit weekly-level variation in the Uber and ASU platforms and plausibly exogenous changes in labor market demand and course-week learning tasks (i.e., course demands) to identify key cross-elasticities.[4]

Our main high-frequency results indicate that an additional hour of coursework is associated with about two-thirds of an hour reduction in time connected to the Uber platform (about 1.3 fewer completed trips). Because students devote fewer hours to coursework than to driving, the short-run opportunity cost is modest (about $41 per week, under 10% of weekly earnings), where the average weekly course load reduces Uber platform time by roughly 1.7 hours, equivalently, a 1% decline in Uber hours for a 10% increase in coursework.[5] In addition, we find that a 10% increase in market-level hourly pay decreases study time for the average Uber-student by only about 1.7%, indicating that study time is relatively unresponsive to short-run changes in labor market conditions. We also show that Uber-students actively reallocate their driving schedules when coursework becomes more demanding, while we find no meaningful evidence that coursework reduces work quality or that driving harms academic performance.

We then turn to students' perceived and realized returns from the ASU–Uber program. Survey respondents expect substantial income gains from completing an ASU Online degree, report a higher likelihood of enrolling in a bachelor's degree in scenarios that combine online learning with flexible work arrangements, and often entered the program after a period of non-enrollment in higher education. Administrative evidence on degree progress shows that among the program's first four entry cohorts, 27.2% of entrants have already graduated and the average entrant has accumulated about 81 ASU credits, roughly two-thirds of the credits required for a bachelor's degree. For context, ASU reports an approximately 24% six-year graduation rate among first-time, full-time students at its online campus, although that

[4]Given that our data come from drivers in a large number of cities, changes in labor demand could be driven, for example, by different social events that may happen in these cities. We show that variations in hotel occupancy rates are consistent predictors of labor market pay at the city-week level, even after including a large set of controls.

[5]Because our identifying variation is within-student and across weeks, these estimates capture short-run, intensive-margin reallocation; they are not directly comparable in magnitude to the cumulative or annual crowd-out measured in the prior literature. Accordingly, the magnitude of this high-frequency response should not be interpreted as evidence that flexibility reduces cumulative crowd-out. Throughout the paper, driving/working time refers to the number of hours that drivers spend connected to the Uber platform.

population and follow-up window differ somewhat from those in our data. Combining these progress measures with our estimated opportunity costs, we estimate that, under a plausible 15% post-graduation wage gain, the average entrant would offset expected opportunity costs after roughly five years.

Our findings indicate that, in this setting, the associated forgone Uber earnings due to coursework are modest, and that students actively use scheduling flexibility to adjust to changing course and labor market demands. Overall, the evidence suggests that pairing flexible work with online learning may help some students combine schooling and employment with modest short-run opportunity costs.

2 Institutional Details: The ASU-Uber Partnership

Beginning in the fall semester of 2018, ASU and Uber initiated a partnership to make higher education more accessible to drivers who meet activity, tenure, and performance requirements on the Uber platform. Under the partnership, Uber covers tuition for drivers in the U.S. who have completed at least 3,000 rides and satisfy ongoing driving and ratings requirements.[6] With the guarantee of tuition coverage, participants can choose among several educational options. The option most relevant for our analysis is enrollment in a fully online undergraduate degree program. Notably, ASU does not distinguish between online and in-person degrees on students' transcripts or diplomas, so the degree option may provide a particularly valuable credential for drivers seeking to combine undergraduate education with flexible work. Alternatively, participants may enroll in shorter (arguably lower-commitment) non-degree options, including entrepreneurship or English-language competency courses, though we do not focus on these programs.

Qualifying Uber participants pursuing a fully online undergraduate degree at ASU have over 140 degree programs to choose from. The courses are offered following the standard university calendar, which includes full-term courses (Session C) and half-term courses that run through the first half (Session A) or second half (Session B) of the term. However, within a course, students generally have considerable flexibility over when they watch lecture

[6]The program also extends to beneficiaries of the drivers, such as a spouse, child, sibling, or parent. However, our analysis focuses only on drivers. Approximately 30% of the program participants are family members of the Uber drivers.

videos and, depending on the course and instructor, some additional flexibility in completing assignments and taking exams. Overall, the flexible structure of these courses is important for our analysis because online enrollment gives students substantial discretion over when to complete coursework, while the academic calendar and assignment deadlines still generate predictable variation in course demands over time.

Although the 3,000-ride threshold may appear substantial, observed driving rates in our sample suggest that many active drivers could reach this threshold within a relatively short period. For example, Uber-students in our sample complete an average of roughly 43 weekly trips when they are not actively enrolled in classes. At that rate, drivers would surpass the necessary 3,000 trips in less than a year and a half, and a more casual driver who completes only 30 trips per week would surpass the qualification in under two years. Once this goal is met, drivers must also have reached either Uber Pro Gold, Platinum, or Diamond status. To first achieve Gold status, a driver must meet a location-specific point goal, have an average star rating of 4.85 or greater, and have a cancellation rate (conditional on trip acceptance) of under 4%.[7] Once one of these different Uber statuses has been attained, the driver must continue to reach their location-specific point total, receive a minimum star rating of 4.75, and maintain a cancellation rate below 10% in order to keep their Uber Pro status. The Uber Pro Gold, Platinum, or Diamond status must be held in three-month periods, such that the activity in the previous three months determines the status in the current period.[8]

3 Data Description

Our analysis combines high-frequency administrative data from ASU and Uber with original survey data and aggregate longer-run educational outcomes. Data provided by ASU enables us to monitor the interactions of Uber-students and their classmates with each course's

[7]The particulars of the point-based goals depend on location, but we can provide some general insights. These points are earned through the completion of rides, with the number of points awarded per completed trip being one or more, contingent upon the time of day and other factors. In most areas (with some exceptions), drivers must accrue between 200 and 600 points within a three-month time frame to sustain their status. As an illustrative example, assuming a midpoint target of 400 points and 1.5 points per trip, our rough calculations suggest that drivers would need to complete less than 21 trips each week to meet this goal. These requirements appear unlikely to bind for most drivers in our sample: full-time drivers (i.e., drivers who average at least 25 hours driving per week when not enrolled in ASU courses) complete about 49 trips each week, on average; part-time drivers complete about 28 trips per week.

[8]The following link provides further details concerning the program's characteristics: https://uber.asu.edu/.

online platform on a weekly basis. We complement the student activity and performance data with Uber administrative data on drivers' weekly activity, including time connected to the platform, completed trips, and earnings, for participants in the ASU-Uber partnership, excluding family-member beneficiaries. These data also include a subset of drivers eligible for the ASU-Uber program who chose not to enroll. Notably, this dataset represents student drivers nationwide, not just those residing in Arizona. Additionally, we have access to data on market-level demand in each driver's local labor market. We merged these two datasets at the individual-week level. For our empirical analysis, we omitted the peak period of the COVID-19 outbreak, as the pandemic significantly disrupted Uber activities. Consequently, our sample spans from August 2018 to March 2020, and from January 2021 to May 2021.[9]

To evaluate program participants' subjective valuation of school and work flexibility, in November 2023, we conducted a survey of (currently enrolled) Uber-students and a random subset of their classmates to augment the data from the ASU and Uber platforms. This survey gathered a host of additional information of interest, including allocation of time in activities beyond the platforms, expectations regarding labor market and graduation outcomes, and preferences for flexible education and learning formats. Unfortunately, the survey data cannot be integrated with the Uber and ASU datasets. Finally, we obtained aggregate graduation and credit-accumulation statistics through December 2025 for the first four cohorts of ASU-Uber program participants, which include the three entry cohorts represented in our main analytic sample. As with the survey, we cannot directly link these outcomes to individuals in our high-frequency analysis.

We next present summary statistics and key features of the ASU and Uber data, and also briefly describe our survey design. We defer discussion of the aggregate longer-run administrative records on credit accumulation and graduation to Section 9. Appendix A provides further details on sample construction and variable definitions.

3.1 ASU Data

The first data set used in our analysis comes from ASU students enrolled in fully online undergraduate programs. Because the degree programs available through the ASU-Uber

[9]We choose to include the Spring 2021 term in our analysis because, by that point, both Uber driving behaviors and ASU Online activity had roughly returned to pre-pandemic levels. However, our primary findings remain qualitatively similar when the analysis is confined to the pre-COVID-19 period.

Table 1: Student-level observables, by ASU-Uber program status

	Uber-Students		Class Peers		Difference
	Mean	Std. Dev.	Mean	Std. Dev.	Mean
Background Characteristics					
Age	39.73	(10.030)	28.51	(8.023)	11.22***
Female	0.15	(0.361)	0.61	(0.489)	-0.45***
White	0.36	(0.480)	0.58	(0.493)	-0.22***
Black	0.33	(0.470)	0.07	(0.256)	0.26***
Hispanic	0.15	(0.353)	0.21	(0.409)	-0.07***
Asian	0.07	(0.251)	0.05	(0.211)	0.02***
First generation	0.57	(0.495)	0.39	(0.489)	0.18***
AZ resident	0.06	(0.238)	0.23	(0.422)	-0.17***
Financial need					
None	0.00	(0.057)	0.01	(0.097)	-0.01***
Very low	0.03	(0.165)	0.06	(0.232)	-0.03***
Low	0.08	(0.264)	0.11	(0.309)	-0.03***
Moderate	0.31	(0.461)	0.35	(0.478)	-0.05***
High	0.09	(0.289)	0.07	(0.250)	0.03***
Very high	0.49	(0.500)	0.39	(0.487)	0.11***
Ever Pell eligible	0.50	(0.500)	0.49	(0.500)	0.01
Academic Profile					
Incoming GPA	2.71	(0.587)	2.88	(0.634)	-0.17***
Transfer status	0.62	(0.486)	0.72	(0.447)	-0.11***
Transfer hours	73.49	(51.525)	52.47	(39.369)	21.02***
Number of terms enrolled	2.60	(1.602)	4.64	(3.280)	-2.04***
Average credit hour load	7.95	(3.027)	8.77	(3.147)	-0.82***
STEM degree	0.39	(0.489)	0.30	(0.456)	0.10***
Observations	1540		64248		

Notes: "Uber-students" refers to ASU students participating in the ASU-Uber partnership. "Class peers" corresponds to all other students enrolled in the same classes as the Uber-students. First-generation refers to students whose parents did not graduate college. "AZ resident" indicates if the student is listed as an Arizona resident. Each of the financial need variables "None", "Very low", "Low", "Moderate", "High", and "Very high" indicate the financial need status of the student, which is constructed by ASU based on students' applications to federal student aid. "Ever Pell eligible" indicates if the student was ever eligible for a Pell grant. The variable incoming GPA corresponds to a student's transfer GPA or, if transfer GPA is inapplicable, high school GPA. "Transfer status" is an indicator variable denoting whether the student transferred from another higher-education institution, and "transfer credit hours" refer to the number of credit hours carried over from their previous institution. "Number of terms enrolled" denotes the number of distinct terms that the student has been enrolled in ASU courses. "Average credit hour load" reflects the student's average number of credit hours per term, and "STEM degree" indicates if the student is in a science, technology, engineering, or math degree program. The sample is restricted to the estimation-sample Uber-students and their class peers in the same sample courses, measured at each student's last term in the sample. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

partnership are operated through ASU Online, students interact with most course materials through Canvas, ASU's online learning management system. This feature allows us to observe high-frequency measures of student engagement at the student-course-week level.

3.1.1 Characterizing ASU-Uber-students

Background Characteristics Table 1 shows background characteristics and proxies for academic preparation of Uber-students and their classmates enrolled in ASU Online undergraduate programs. The first set of columns reports means and standard deviations for

Table 2: Weekly Canvas activity and outcomes, by sample

	(1)		(2)		(3)		(4)	(5)
	Uber-Students Active in Uber		Uber-Students Inactive in Uber		Class Peers		Col (1) - (3)	Col (1) - (2)
	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.	Difference	Difference
Activity								
Days connected to Canvas per course	3.57	(2.30)	3.75	(2.27)	3.48	(2.12)	0.09***	-0.18***
Clicks per course	75.97	(89.33)	79.04	(85.57)	63.94	(69.70)	12.03***	-3.07***
Hours online per course	1.57	(2.12)	1.61	(2.10)	1.25	(1.63)	0.31***	-0.04**
Outcomes								
Assignments submitted per course	1.97	(4.14)	1.99	(2.92)	2.12	(4.05)	-0.15***	-0.02
Assignment share per course (%)	90.19	(27.59)	90.55	(26.81)	91.56	(25.10)	-1.37***	-0.36
Point share per course (%)	79.40	(29.31)	80.13	(28.66)	81.16	(28.64)	-1.76***	-0.74**
Observations	43527		12201		1831331			

Notes: All statistics reported in the table are at the student-by-course-by-week level. "Uber-Students: Active in Uber" denotes course-weeks in which the student completes at least one trip and is enrolled in ASU classes. "Uber-students: Inactive in Uber" denotes weeks in which the student completes no trips but is enrolled in ASU classes. "Class peers" correspond to all other students who are enrolled in the same classes as the Uber-students. Assignment and point shares refer to the total number of assignments submitted and points earned relative to the number of assignments due and points possible, respectively. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

Uber-students, while the second reports the same statistics for their classmates who are not ASU-Uber program participants. Throughout the remainder of the text, we denote this group of students as "class peers". Uber-students differ in multiple dimensions from their classmates. They are, on average, 11 years older, overwhelmingly male (85%), and more racially diverse. For instance, 33% of the Uber-students are Black, whereas the analogous share for their peer population is only 7%. In addition, Uber-students are more likely to be first-generation (i.e., do not have a parent who graduated from college) and to reside outside Arizona. The middle panel shows that Uber-students have higher financial needs (based on FAFSA records): 49% of them are designated as "very high" financial need versus 39% of their peers. The last panel shows that Uber-students have lower incoming GPAs than their counterparts, are less likely to enter ASU as transfer students, but have accumulated more transfer credit hours on average. They have also completed fewer terms at ASU, are more likely to pursue a major in STEM, and have a lower average credit-hour load per term than their class peers. Note that most ASU Online courses that we consider in our analysis are worth 3 credit hours, so Uber-students and their class peers complete roughly 2.7 and 2.9 courses per term, respectively.

Weekly Academic Engagement Table 2 shows proxy measures for student engagement (i.e., active days, clicks, hours online) and performance in ASU Online courses at the student-

course-week level. The sample is divided into three groups: Uber-students active in Uber, Uber-students inactive in Uber, and class peers, where active and inactive distinguish the weeks in which Uber-students were actively participating on the Uber platform; if they complete at least one trip during a week, they are classified as active for that week. On average, Uber-students show greater engagement on Canvas than their classmates. They tend to spend more days per week connected to the ASU platform, record more clicks, and spend more time on it.[10] Uber-students also slightly increase their engagement on Canvas during weeks when they are inactive on the Uber platform. Finally, Uber-students, regardless of Uber activity status, spend about 1.6 hours online per active course in a given week. Because courses are distributed across Sessions A, B, and C, students are enrolled in approximately 1.9 active courses in an average enrolled week, yielding roughly 3 hours of online study activity per week across all courses.[11]

In terms of weekly academic performance, Uber-students submit, on average, 1.97 assignments per course during active driving weeks, slightly less than their peers (2.12), representing 90.2% of the total assignments requested. They obtain approximately 80% of the total points, which is similar to the point share of their class peers (81%).[12] Finally, differences in weekly academic performance between active and inactive Uber weeks are small.

A possible concern with these data is that our various Canvas engagement variables may be incomplete measures of study time and effort. However, the students' self-reported

[10]Appendix Table G.1 shows that our measures of online activity (i.e., days connected to Canvas, number of logins and clicks, and time spent on the platform) are highly correlated. In particular, the high correlation between hours online and clicks suggests that idle time is not likely to be an important concern when proxying study time with "hours online".

[11]Because direct measures of study time are rare, it is difficult to benchmark whether observed Canvas time is high or low. Using course evaluation surveys, Ahn et al. (2024) find that students at the University of Kentucky spend, on average, 2.4 hours per week studying for a non-STEM course and 3.5 hours per week studying for a STEM course. Babcock and Marks (2011) document that U.S. college students' average weekly study time, including class time, declined from 40 hours in 1961 to 27 hours in 2003. These benchmarks suggest that our Canvas-based measure captures only part of total academic effort, which may also include reading, studying, or completing work off platform.

[12]The point share is determined by dividing the points earned in a course for a given week by the points possible for the student in a given course-week. Assignments submitted refer to the number of assignments the student submitted throughout the week (assignments do not necessarily need to be submitted in the week they are due). In Appendix Table G.2, we provide an alternative version of Table 2 where we re-weight the control group by coarsened exact matching (see Iacus et al., 2012) on many of the observable characteristics presented in Table 1. After doing so, the Uber-students' Canvas activity measures are much more similar to the control group, suggesting that once we control for fixed observable characteristics, the Uber-students' study behaviors are quite similar to their non-Uber peers.

Table 3: Course-level summary statistics, by sample

	(1)		(2)		(3)
	Uber Students		Class Peers		Col (1) - (2)
	Mean	Std. Dev.	Mean	Std. Dev.	Mean
Activity					
Total hours on Canvas	14.63	(12.73)	11.47	(9.62)	3.16***
Assignments submitted	18.38	(16.89)	19.37	(17.54)	-1.00***
Outcomes					
Passed course	0.76	(0.43)	0.83	(0.38)	-0.07***
Numeric course grade	2.49	(1.66)	2.70	(1.54)	-0.21***
Observations	6591		213842		

Notes: "Uber-Students" denotes ASU students who participate in the ASU-Uber partnership. "Class peers" corresponds to all other students that are enrolled in the same classes as the Uber-students.*, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

study time allocation on and off the platforms (see subsection 3.3) and the strong correlation between Canvas hours and academic performance (discussed in Section 8) suggest that our measures of study effort constitute good proxies for actual study time. Furthermore, because our empirical specifications include individual fixed effects, they account for time-invariant differences across students in the extent to which Canvas activity captures total study effort. Remaining concerns would arise if within-student changes in off-platform study time are systematically correlated with weekly Uber activity.

Course-Level Academic Outcomes Table 3 reports course-level measures of Canvas activity and academic outcomes for Uber-students and their class peers. The top panel summarizes total activity over the course, while the bottom panel reports course completion and grade outcomes. On average, Uber-students spend approximately 14.6 hours on Canvas per course, compared with 11.5 hours among their class peers. For reference, most courses last seven to eight weeks, though a small number of courses run for 15–16 weeks. Uber-students submit roughly 18.4 assignments per course, whereas their classmates submit approximately 19.4. These patterns suggest that Uber-students remain actively engaged with course materials, despite the additional work demands they may face outside of school.

The bottom panel of Table 3 shows that Uber-students have a 76% passing rate and average course grades of 2.49. Unlike activity, these measures of academic performance are somewhat worse compared to their peers (i.e., a passing rate of 83% and an average grade of 2.70).[13] However, these differences are somewhat expected given that enrolled Uber-

[13]Appendix Table G.3 also presents summary statistics regarding the progress of Uber-students and their

students, on average, are from less advantaged backgrounds and have substantially lower prior academic credentials. In light of the evidence presented in this table, the higher levels of engagement presented in Table 2 may reflect greater effort by Uber-students to keep pace with course material.

Overall, the main takeaway from these summary statistics is that Uber-students, despite coming from more disadvantaged backgrounds than their class peers, have only slightly lower course completion rates and grades but somewhat higher levels of engagement in learning activities than their classmates.

3.1.2 Within-Term Variation in Canvas Engagement

To further characterize within-term variation in course effort, we estimate a descriptive event-time specification for Uber-students' weekly Canvas hours. Specifically, we analyze how study time in the different weeks of the term compares to the third week before the beginning of the academic semester.[14]

The results for eight-week courses, the modal course length in our sample, are presented in Figure 1. As expected, Canvas hours are essentially zero in the weeks leading up to the

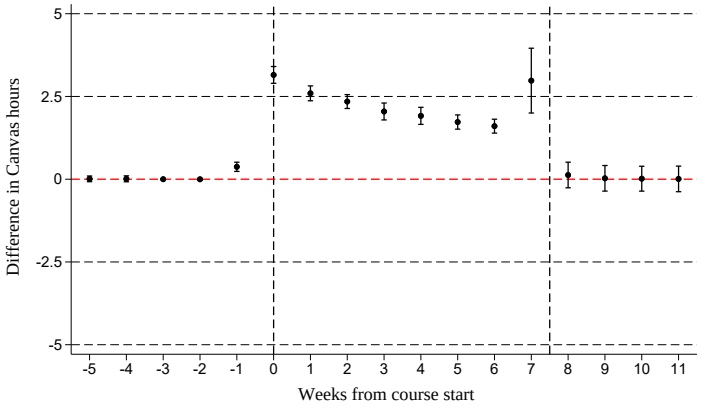
class peers (i.e., cumulative GPA, and credits to date). Additionally, Appendix Table G.4 presents the same summary statistics as in Table 3, but when we re-weight the "Class Peers" to be more similar to the Uber-students on observables. Upon re-weighting, the differences in activity across groups attenuate considerably.

[14]We estimate a specification of the following form:

$$CourseE_{i,c,t} = \sum_{\substack{y=-5 \\ y\neq -3}}^{T} \alpha_y \mathbb{I}(t - t_c^* = y) + \beta_1 CourseWork_{i,t,-c} + \beta_2 MarketHourlyPay_{l,t} + \beta_3 Weather_{l,t} + X_{i,t}\Phi + \psi_i + \delta_m + \phi_c + \gamma_l + \varepsilon_{i,c,t}, \quad (1)$$

where $CourseE_{i,c,t}$ is Canvas hours for Uber-student i, taking course c, in week t. $CourseWork_{i,t,-c}$ captures course peer activities in the other courses $(-c)$ in which the student is enrolled (controlling or not for this variable provides very similar results). The vector $X_{i,t}$ includes controls for the number of course days in the week and an indicator for whether the week includes Christmas. ψ_i, δ_m, ϕ_c, and γ_l denote individual, month-year, classroom (e.g., a specific section of ECN 101), and city fixed effects, respectively. $MarketHourlyPay_{l,t}$ and $Weather_{l,t}$ denote the average market-level hourly pay and second-order polynomials of rainfall and snowfall, respectively, in the labor market l for individual i in week t. $\varepsilon_{i,c,t}$ corresponds to the idiosyncratic shock. Finally, $\mathbb{I}(t - t_c^* = y)$ denotes indicators for the week of the course, y, where t_c^* is the first week of the course ($y = 0$). The excluded week is the week that is three weeks prior to the start of the course. Thus, each $\hat{\alpha}_y$ captures average Canvas hours in event week y, relative to the third week before the course starts, conditional on the included controls and fixed effects. Section 5 discusses the rationale for this specification. To appropriately capture the evolution of driving hours and study hours, we exclude individuals who drop out or withdraw at some point during the course.

Figure 1: Variation in study time during eight-week courses



Notes: OLS coefficient estimates from a version of Equation (1) where the dependent variable is hours on Canvas and 95% confidence intervals are plotted. The estimating sample excludes individuals who appear to have dropped out of the course. Standard errors are clustered at the classroom level.

beginning of the course.[15] Canvas hours increase sharply to roughly three hours per week in the first week of class, decline gradually over the middle of the course, and then rise again in the final week. This pattern suggests that students devote substantial time to Canvas at the start of the course, likely as they orient themselves to course requirements and initial assignments, before reducing weekly engagement during the middle of the term. The final-week increase likely reflects end-of-course assignments, exams, or other culminating course requirements.

3.2 Uber Data

The Uber driver data include background characteristics and individual weekly driving activity corresponding to two groups of drivers: Uber-students and a random sample of drivers who qualify to participate in the ASU-Uber partnership program but did not enroll (henceforth, Uber peers). Additionally, we have access to city-level labor market characteristics and weather conditions.[16]

3.2.1 Characterizing ASU-Uber Drivers

Background Characteristics Table 4 shows how Uber-students compare with Uber peers; Uber peers are drivers who qualify for the ASU-Uber program but do not enroll

[15]The point estimate for week -1 is slightly above zero because some courses allow students to access the Canvas page before the official start of the term. This is why we use week -3 as the omitted category.

[16]The drivers in our sample work in 111 distinct geographic regions/cities across the United States. Information on weather conditions was downloaded from the PRISM Climate Group at Oregon State University.

Table 4: Driver-level observables, by sample

	(1)		(2)		(3)
	Uber-Students		Uber Peers		Col (1) - (2)
	Mean	Std. Dev.	Mean	Std. Dev.	Mean
Background Characteristics					
Female	0.15	(0.35)	0.10	(0.30)	0.05***
Age†	42.62	(10.59)	49.67	(12.80)	-7.05***
English	0.99	(0.07)	0.91	(0.29)	0.09***
Spanish	0.00	(0.05)	0.07	(0.25)	-0.07***
Ever driven in AZ	0.06	(0.24)	0.02	(0.15)	0.04***
Uber activity					
Weeks in Uber data	128.91	(27.57)	111.75	(37.76)	17.16***
Active weeks in Uber data (%)	0.79	(0.16)	0.81	(0.17)	-0.02***
Lifetime completed trips	8118.40	(4554.43)	7798.29	(4732.58)	320.12***
Lifetime average rating	4.92	(0.04)	4.90	(0.05)	0.02***
Observations	1540		96513		

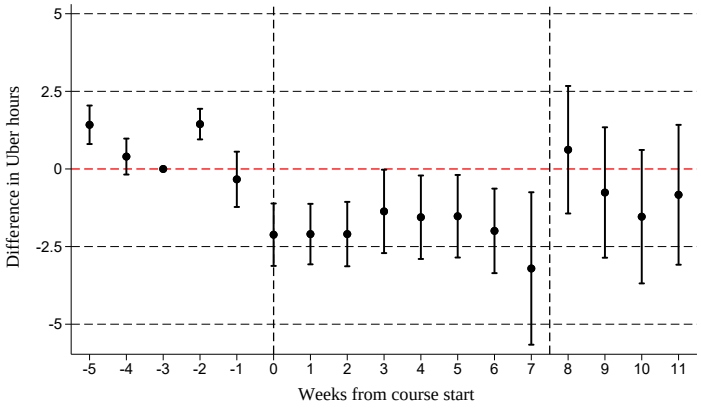
Notes: "Uber-students" denotes ASU students who participate in the ASU-Uber partnership. "Uber peers" denotes drivers who qualify for the ASU-Uber partnership but do not enroll in ASU courses. "Active weeks" refer to weeks in which driving hours are non-zero, and "lifetime" refers to the driver's full history on the Uber platform. "Lifetime average rating" ranges from zero to five. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.
†Age is based on decade of birth rather than exact year of birth, as this is how Uber released the information. English and Spanish refer to the driver's primary language.

in ASU courses. Uber-students are, on average, more likely to be female and younger than their Uber peers. Uber-students are also more likely to report English as their primary language, less likely to report Spanish as their primary language, and more likely to work in Arizona than their driving counterparts (i.e., 6% vs. 2%). The bottom panel of this table also indicates that Uber-students have longer Uber tenures than their peers. In particular, they have been driving for more than 128 weeks, and during 79% of those weeks (i.e., 102 weeks), they have been active on the Uber platform. On average, they have completed more than 8,100 trips, indicating an important degree of attachment to driving with Uber. Finally, Uber-students have, on average, high Uber ratings consistent with the eligibility requirements of the ASU-Uber program.

Weekly Driving Engagement Appendix Table G.5 reports weekly summary statistics on Uber driving activity during weeks when Uber-students are enrolled in ASU classes. The top panel shows that Uber-students spend, on average, 18.25 hours per week on the Uber platform and complete 36.62 trips.[17] The bottom panel shows that Uber-students earn roughly $442 per week, corresponding to average earnings of $12.37 per trip and $24.26 per

[17]This measure of Uber hours includes both active time (i.e., driving to pick up or complete a ride) and inactive time (i.e., waiting for a pickup request). In the individual-level data, we observe only total time on the platform.

Figure 2: Variation in driving hours during eight-week courses



Notes: OLS coefficient estimates from a version of Equation (1) where the dependent variable is hours on the Uber platform and 95% confidence intervals are plotted. The estimating sample excludes individuals who appear to have dropped out of the course. Standard errors are clustered at the classroom level.

hour. The final rows of Table G.5 show that tips account for approximately 10% of Uber-students' earnings per trip and per hour, and that these drivers maintain very high average ratings.

Appendix Table G.6 reports correlations between Uber activity measures—including completed trips and time on the platform—and measures of total pay, incentive pay, and tips. Hours on the platform and completed trips are highly correlated, suggesting that total platform time is a useful proxy for driving engagement, even though it includes some inactive waiting time.

3.2.2 Variation in Weekly Uber Driving

As with study time, we complement the summary statistics in Table G.5 by examining how driving hours evolve over the course of the term. We estimate a descriptive event-time specification in which the omitted category is the third week before the course begins.[18] Figure 2 plots the estimated event-week coefficients for driving hours in eight-week courses.

Driving hours decline slightly in the week immediately before the course begins and fall by roughly two hours per week once the course starts. This decline remains fairly stable over the term, with a somewhat larger reduction in the final week of the course. The final-week decline likely reflects final exams, end-of-course assignments, or other culminating course

[18]The empirical specification is analogous to Equation (1), but where the dependent variable is hours on the Uber platform.

requirements. After the course ends, driving hours are no longer statistically distinguishable from pre-course levels.

Although Figures 1 and 2 suggest offsetting movements in Canvas engagement and Uber driving, we do not interpret these descriptive patterns as evidence of hour-for-hour substitution between work and study time. Rather, they are informative about short-run, within-semester reallocations that occur while Uber-students are enrolled in ASU Online courses, and should not be interpreted as cumulative year-level crowd-out of work or schooling.

3.3 Survey Data

All participants of the ASU-Uber program enrolled in Arizona State University's online courses in November 2023 received an email invitation to participate in our survey. A randomly selected, equal number of their course peers also received an email invitation to participate. The survey aimed to collect data on students' demographic backgrounds, previous academic experiences, expectations for college and labor market outcomes, time allocation outside the online platforms, and their willingness to pay for flexible learning and working options. As an incentive for participation, respondents entered into a lottery, offering a chance to win one of 250 $15 Amazon gift cards or one of 10 $100 Amazon gift cards. Out of 5,478 individuals surveyed, 692 (13%) responded. The respondents included 243 Uber-students, 105 family members of Uber drivers participating in the program, and 344 class peers. Because our analysis does not focus on family members, their responses are excluded. The median time taken to complete the survey was 22 minutes.

Appendix Table G.7 presents a comparison of the demographics and field of study between Uber-students (excluding family members) in our survey sample and those recorded in the administrative data (i.e., platform data). This comparative analysis reveals that the demographic characteristics of the two groups are broadly similar.[19] The primary deviation is observed in the female participation rate. However, this discrepancy arises because the survey also encompasses Uber Eats drivers, representing a more gender-balanced group.

The survey also sought to determine the proportion of study time Uber-students spend on the platforms and whether they drive for other ride-sharing services. Survey results reveal

[19]Appendix Figure H.1 shows that approximately half of Uber-students were either not enrolled or on a break from another institution prior to joining ASU. In addition, approximately half of the Uber-students who were previously enrolled or taking a leave from college were attending 2-year institutions.

that 77% of the Uber-students' study time is conducted on the Canvas platform, and they dedicate over 83% of their ride-sharing hours to Uber. These shares are consistent with Hyman et al. (2020), which shows that less than one-third of drivers use both the Uber and Lyft platforms in Seattle. These responses suggest that the ASU and Uber platform data capture a substantial share of students' academic engagement and ride-share activity. Nevertheless, the platform data do not capture all off-platform study time or all sources of income, so our measures may understate students' total academic effort and economic activity.

4 Analytical Framework

In this section, we present a simple framework in which students allocate effort between work and coursework in an environment where both labor-market incentives and academic demands vary across weeks. The model is intentionally stylized, but it clarifies why our empirical specifications focus on exogenous shifts in labor-market opportunities and coursework demands rather than treating work and study effort as direct causal inputs into one another.

Agent's Maximization Problem Equation (2) poses the problem of the agent. Individual i chooses weekly (t) work effort, e_{ilt}, and study effort, e_{ict}, to maximize a weighted sum of earnings and academic performance net of effort costs. Earnings equal $w_t e_{ilt}$, while grades depend on study effort, ability (Ab_i), and coursework difficulty (ς_t), as specified in Equation (3). The parameters λ_l and λ_c govern the costs of work and study effort, and λ_{lc} allows the marginal cost of either activity to increase with effort devoted to the other. Finally, the additional constraints reflect basic and intuitive restrictions on α and the λs. Given these preferences, the problem of the agent is to choose the optimal vectors of effort in each activity $(\mathbf{e}^*_{ic}, \mathbf{e}^*_{il})$.[20]

[20] As mentioned earlier, the goal of the model is to provide a framework to guide the empirical specification. Thus, the model is simplified along several dimensions. For example, the decision of each period is treated as static in nature. Similarly, Uber earnings are taken as given, and we do not allow for the possibility that individuals may strategically decide to drive at times within the week with higher rates. The last simplification is partly also made because we only observe Uber earnings at the weekly level.

$$\max_{\mathbf{e}_{il},\mathbf{e}_{ic}} \sum_{t=1}^{T} \Big(\alpha w_t e_{ilt} + (1-\alpha) G_{it} \Big) - \lambda_l \sum_{t=1}^{T} e_{ilt}^2 - \lambda_c \sum_{t=1}^{T} e_{ict}^2 - \underbrace{\lambda_{lc} \sum_{t=1}^{T} e_{ilt} e_{ict}}_{\substack{\text{Negative complementarities of effort} \\ \text{between work and college activities}}} , \quad (2)$$

where

$$G_{it} = \beta_0 + \beta_1 e_{ict} + \beta_2 Ab_i + \beta_3 e_{ict} Ab_i + \beta_4 \varsigma_t + \beta_5 e_{ict} \varsigma_t + \varepsilon_{it} \quad (3)$$

$$\alpha \in [0,1]$$

$$\lambda_l \geq 0, \lambda_c \geq 0, \lambda_{lc} \geq 0, 4\lambda_l \lambda_c > \lambda_{lc}^2$$

Optimal Effort To characterize the model's comparative statics, we focus on the interior solution in which both work and study effort are positive. This abstracts from the lower-bound constraint that effort cannot be negative, but it allows us to derive how optimal effort responds to changes in wages and coursework demands. The first-order conditions of the agent's problem imply that the optimal levels of work and study effort satisfy:

$$e_{ilt}^* = \frac{(\alpha-1)\lambda_{lc}[\beta_1 + \beta_3 Ab_i + \beta_5 \varsigma_t] + 2\alpha\lambda_c w_t}{4\lambda_c\lambda_l - \lambda_{lc}^2} \quad (4)$$

$$e_{ict}^* = \frac{2(1-\alpha)\lambda_l[\beta_1 + \beta_3 Ab_i + \beta_5 \varsigma_t] - \alpha\lambda_{lc} w_t}{4\lambda_c\lambda_l - \lambda_{lc}^2}. \quad (5)$$

Equations (4) and (5) imply that higher wages increase work effort and reduce study effort, while greater coursework demands increase study effort and reduce work effort. They also show that work and study effort are jointly chosen outcomes responding to underlying incentives. This motivates our empirical focus on exogenous variation in wages and coursework demands rather than specifications that treat realized work and study effort as direct causal determinants of one another.[21]

5 Empirical Strategy

Our empirical strategy exploits week-to-week variation in course demand and labor market conditions. Thus, the coefficients in our main specifications should be interpreted as within-

[21] Wooldridge (2015) similarly cautions against modeling simultaneously chosen work and study hours as direct determinants of one another.

term, intensive-margin substitution responses. The estimates can therefore be interpreted as short-run, intensive-margin responses in work and study effort, with the wage response corresponding to a Frisch-type labor-supply response. In other words, they characterize how students reallocate time when both work and study effort can be adjusted at a high frequency. These estimates are informative about short-run displacement and strategic scheduling, but they do not identify the cumulative annual effect of enrollment relative to non-enrollment, nor the overall long-run crowd-out of work by schooling or schooling by work.

5.1 The Impact of Learning Activities on Labor Market Supply

We use weekly individual-level data on Uber driving and ASU course engagement to estimate how coursework affects labor supply. Motivated by Equation (4), our primary specification is:

$$\begin{aligned} WorkE_{i,t} = \beta_0 + \beta_1 \sum_{c \in C_{it}} CourseWork_{c,t} + \beta_2 MarketHourlyPay_{l(i,t),t} \\ + \beta_3 Weather_{l(i,t),t} + \psi_i + \delta_m + \phi_b + \gamma_{l(i,t)} + \epsilon_{i,t}, \end{aligned} \tag{6}$$

where $WorkE_{i,t}$ denotes Uber labor supply, measured using either total platform hours or completed trips. $CourseWork_{c,t}$ measures weekly course demand using the average Canvas hours of non-Uber peers in course c, summed across the courses in which student i is enrolled. $MarketHourlyPay_{l(i,t),t}$ is average active hourly pay among eligible non-enrolled drivers in the student's labor market, and $Weather_{l(i,t),t}$ includes second-order polynomials in local rainfall and snowfall. The specification also includes individual (ψ_i), month-year (δ_m), course-bundle (ϕ_b), and city/labor market ($\gamma_{l(i,t)}$) fixed effects. Finally, $\epsilon_{i,t}$ represents an idiosyncratic shock.[22]

The coefficient of interest is β_1, which captures how weekly coursework demand affects Uber labor supply among enrolled student-drivers. The identifying assumption is that, conditional on individual fixed effects, course-bundle fixed effects, month-year fixed effects, market-level pay, and weather, week-to-week variation in non-Uber peers' Canvas activity represents exogenous variation in coursework demand faced by Uber-students.[23] Since iden-

[22] Appendix A.1 provides additional details on variable construction. Results are robust to defining market hourly pay using total platform time rather than active driving time.

[23] Because ASU Online courses are asynchronous and students generally do not know one another or live

tification comes from within-student variation during enrolled terms, the estimates are less likely to be confounded by time-invariant differences across students or by pre-enrollment differences in attachment to school or work.[24] We also estimate specifications with city-week fixed effects, which absorb local labor-market shocks more flexibly but preclude separate identification of β_2.

The coefficient β_2 captures the responsiveness of Uber labor supply to transitory changes in local hourly pay. Identifying this parameter requires that unobserved city-week shocks to Uber pay are not otherwise related to individual labor supply after conditioning on the included controls and fixed effects. Weather is a natural concern because it may affect both ride demand and the cost of driving, so all baseline specifications include second-order polynomials in rainfall and snowfall. We also provide supporting evidence in Appendix B that market-level hourly pay responds to local demand shocks, using weekly hotel occupancy data. In particular, a one-standard-deviation increase in local hotel occupancy is associated with a 0.25 standard deviation increase in market-level hourly pay.

5.2 The Impact of Labor Market Conditions on College Effort

We next estimate how transitory labor market conditions affect course engagement. As in the labor-supply specification, our estimating equation is motivated by the optimal study-effort condition in Equation (5). The primary specification is:

$$\begin{aligned} CourseE_{i,c,t} = \delta_0 + \delta_1 CourseWork_{i,t,c} + \delta_2 \sum_{\substack{c' \in C_{it} \\ c' \neq c}} CourseWork_{i,t,c'} \\ +\delta_3 MarketHourlyPay_{l(i,t),t} + \delta_4 Weather_{l(i,t),t} + \delta_5 X_{c,t} + \pi_i + \alpha_m + \rho_c + \eta_{i,c,t}, \end{aligned} \tag{7}$$

where $CourseE_{i,c,t}$ denotes course-level effort of student i on class c during week t. While Equation (6) is estimated at the individual-week level, this specification is estimated at the individual-course-week-level. We measure course effort using weekly Canvas hours, clicks, or logins in course c. $CourseWork_{i,t,c}$ captures course demand in course c, measured using the average Canvas activity of class peers who are not ASU-Uber program participants. The

in the same cities, peer collaboration in coursework is likely limited.

[24] Though not a threat to identification per se, including student-drivers who have dropped out of their courses in our sample could complicate our interpretations of $\hat{\beta}_1$ and $\hat{\beta}_2$. Appendix A.2 describes how we identify likely course dropouts and shows that excluding them has little effect on the main estimates.

summation over $CourseWork_{i,t,c'}$ with $c' \neq c$ captures coursework demand in the student's other enrolled courses during the same week. $MarketHourlyPay_{l(i,t),t}$ and $Weather_{l(i,t),t}$ are defined as in Equation (6). $X_{c,t}$ denotes number of course days in the week and π_i, α_m, and ρ_c denote individual, month-year, and classroom fixed effects, respectively.[25] A classroom is defined as a given course taught by a given instructor in a given term. Finally, $\eta_{i,c,t}$ corresponds to the idiosyncratic shock.

The parameter of interest δ_3 indicates how sensitive college effort is to labor market conditions. The identifying assumption is that, conditional on the fixed effects, observable controls, and weather conditions, market-level hourly pay is unrelated to unobserved shocks to an individual student's course engagement. As a robustness check, Section 6.2 presents a placebo test using ASU Online students who are not enrolled in the ASU-Uber program to test for the validity of our identifying assumption.

5.3 The Effects of Work and Study Effort on Academic Performance

Finally, we examine how work and study effort affect weekly performance and final course grades. We first estimate:

$$PointShare_{i,c,t} = \pi_0 + \pi_1 WorkHours_{i,t} + \pi_2 Weather_{l(i,t),t} + \pi_3 X_{i,c,t} + \psi_i + \delta_m + \rho_c + \varepsilon_{i,c,t}, \quad (8)$$

where $PointShare_{i,c,t}$ refers to the share of possible points, on a 0–100 scale, that student i earned in course c during week t and $WorkHours_{i,t}$ is total time on the Uber platform. $Weather_{l(i,t),t}$ denotes the standard set of weather controls, and $X_{i,c,t}$ includes controls for the number of course days in the week and an indicator for whether there were no possible points to earn in that course-week.[26] ψ_i, δ_m, and ρ_c denote individual, month-year, and classroom fixed-effects, respectively. We also estimate analogous specifications in which Canvas hours replace Uber hours as the measure of weekly effort.

In both cases, we focus on estimates from instrumental variables (IV) specifications that attempt to account for potential simultaneity between work or study effort and weekly course

[25] We include the number of course days in the week as a control because some weeks of class (such as the first and final weeks of the course) do not start on Monday or end on Friday. As a result, Canvas activity is, in a sense, mechanically lower due to fewer days of potential activity.

[26] We code weeks with no possible points as receiving the maximum share, but the results are robust to excluding those weeks, or coding them as the minimum share.

outcomes. We instrument an individual's weekly driving hours with the average market-level hourly pay in that week for drivers who were eligible for the ASU-Uber program but did not enroll. The intuition of the instrument is that, after conditioning on observable controls and fixed effects, market-level earnings should induce variation in a given student's driving time but are otherwise exogenous to course outcomes. Similarly, we instrument an individual student's total Canvas time in a given week with the average total Canvas hours of their non-Uber peers in that week. The key identifying assumption in this case is that, conditional on the fixed effects and observable characteristics, peer Canvas time captures exogenous week-to-week variation in course workload on the platform, and is unrelated to other idiosyncratic shocks that might affect an individual Uber student's performance in a given week.

Second, we examine how total Uber driving during a course relates to final course grades:

$$CourseG_{i,c} = \gamma_0 + \gamma_1 \sum_{t \in \mathcal{T}_c} WorkHours_{i,t} + \rho_c + \delta_{i,term} + \eta_{i,c}, \tag{9}$$

where $CourseG_{i,c}$ denotes student i's final grade in course c. $\mathcal{T}_c$ denotes the weeks spanned by course c, so the summation of $WorkHours_{i,t}$ denotes the total hours driven for Uber throughout the entire course. The specification includes classroom and individual-by-term fixed effects, so identification comes from differences in driving exposure across courses taken by the same student in the same term. For work hours, this variation arises from courses offered in different sessions, since concurrent courses span the same weeks and therefore share the same realized driving hours.

A key threat to identification in the model presented in equation (9) arises if work hours are endogenous to a student's particular ability in a course (i.e., courses in which a student is particularly capable may have high driving hours and high achievement). To address this concern, we instrument total Uber hours with average market-level hourly pay over the weeks spanned by the course. This instrument isolates variation in course-level driving time induced by weekly changes in labor market conditions, while remaining plausibly exogenous to overall course outcomes.

Third, we estimate an analogous version of Equation (9) that replaces work effort with

total Canvas hours during the course.[27] By doing so, we can combine the estimates on college effort from this specification with those obtained from Equation (7) to provide a back-of-the-envelope calculation of the effect of labor market hourly pay on academic performance through its impact on course effort. As with work effort, study time is likely endogenous. Again, we account for this by instrumenting an individual student's total Canvas time throughout the course with the average total Canvas hours of their non-Uber peers. Note that, unlike work hours, course-specific Canvas hours can vary across concurrent courses, so identification in the study-effort specification comes from within-term differences across both concurrent and non-concurrent courses.

6 Results

6.1 Coursework and Labor Market Supply

Table 5 presents estimates of Equation (6), which examines how weekly coursework demand affects Uber labor supply. Column (1) indicates that each additional hour of peer coursework is associated with about a 0.68-hour (roughly 41-minute) reduction in weekly time connected to the Uber platform, and Column (4) shows a corresponding reduction of about 1.3 completed trips. Aggregated over the average weekly course load, the estimates imply that coursework reduces Uber platform time by approximately 1.7 hours per week; equivalently, a 10% increase in coursework reduces Uber hours by roughly 1%.[28]

Coupled with the fact that Uber-students earn roughly $24 per hour, we interpret these findings as suggesting that the average short-run opportunity cost of college coursework is approximately $41 per week, or $180 per month, during enrolled weeks (or $21 per week per course enrolled). Columns (2)-(3) and (5)-(6) show that our estimates are robust to controlling for weather conditions and replacing month-year fixed effects with city-week fixed effects.[29]

[27] College effort is a mediating variable for the effect of work effort on academic performance. Therefore, the effect of work effort on academic performance, once conditioning on study effort, is expected to be close to zero.

[28] This value is obtained as $-0.680 \times 2.45 = -1.67$, where 2.45 is the average sum of peer Canvas hours across the courses in which an Uber-student is enrolled. At the course level, the analogous calculation is $-0.680 \times 1.27 = -0.86$ hours per course. Projecting peers' Canvas hours onto Uber-students' actual Canvas hours yields a coefficient close to one.

[29] The number of observations varies slightly across columns because weather data are not available for some cities. Specifications with city-week fixed effects also drop city-weeks with only one observation.

In addition, given the city-week fixed-effects included in our specifications for Columns (3) and (6), we are

Table 5: Determinants of work effort

	(1) Work Hours	(2) Work Hours	(3) Work Hours	(4) Completed Trips	(5) Completed Trips	(6) Completed Trips
Sum of avg. study hours across all c, peers	-0.680***	-0.678***	-0.667***	-1.283***	-1.279***	-1.311***
	(0.071)	(0.072)	(0.100)	(0.146)	(0.146)	(0.204)
Market level active hourly pay, currently elig.	0.122**	0.132***	–	0.351***	0.381***	–
	(0.051)	(0.051)		(0.112)	(0.113)	
Obs.	27475	27404	26265	27475	27404	26265
Individual Fixed Effects	✓	✓	✓	✓	✓	✓
Course Bundle Fixed Effects	✓	✓	✓	✓	✓	✓
Month-Year Fixed effects	✓	✓		✓	✓	
City Fixed Effects	✓	✓		✓	✓	
Weather controls		✓			✓	
City-Week FE			✓			✓
Mean dep. var.	18.24	18.21	18.39	36.52	36.48	36.78
Market level active hourly pay, peers	33.62	33.61		33.62	33.61	
Sum avg. study hours all c, peers	2.45	2.45	2.46	2.45	2.45	2.46
Mean hourly earnings	24.23	24.22	24.38	24.23	24.22	24.38
Mean pay-per-trip	12.37	12.37	12.46	12.37	12.37	12.46
Elasticity w.r.t. peer coursework	-0.091	-0.091	-0.089	-0.086	-0.086	-0.088
Elasticity w.r.t. peer earnings	0.225	0.244		0.323	0.351	

Notes: The dependent variables for Columns (1) to (3) and Columns (4) to (6) are weekly work hours and weekly completed trips, respectively. The variable "Sum of avg. study hours across all c, peers" denotes the sum of the average hours per week that peers spent on Canvas across all courses. The variable "Market level active hourly pay, currently elig." denotes the average active hourly pay in a given week of drivers who are currently eligible for the ASU-Uber partnership, but not enrolled. Specifications include multiple sets of fixed effects. Course bundle fixed effects denote the set of courses taken in a given term. Weather controls include second-order polynomials in total weekly rainfall and snowfall. Elasticities are calculated by multiplying the relevant coefficient estimates by the ratio of mean peer coursework or market-level hourly earnings to the mean of the dependent variable. Standard errors are clustered at the individual level. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

Turning to labor market conditions, the estimates in Table 5 (which should be interpreted with some caution per the discussion in Section 5.1) show that a $1 increase in average hourly pay increases average weekly Uber hours by 0.122 hours, or about 7.3 minutes, and completed trips by 0.35 trips per week.[30] If we consider that the average *active* hourly pay of drivers that do not attend ASU is approximately $33.60, and that the average weekly driving time among Uber-students is 18.24 hours, then a 10% increase in average hourly pay leads to an increase in driving time (trips) of roughly 2.3% (3.2%). Caldwell and Oehlsen (2023) find that the Frisch elasticity for male Uber drivers (our sample is overwhelmingly male) is 0.4,

somewhat limiting the level of identifying variation available. Reassuringly, though, we see that there are, on average, 26 observations in each city-week (ranging from 2 to 73 observations per city-week).

[30]The top sub-figures of Appendix Figure H.2 present the distribution of the labor supply elasticity with respect to coursework and earnings (i.e., calculated based on the average coursework and market-level hourly earnings of each Uber-student).

which is higher than the elasticity we estimate. However, two reasons could explain the differences in findings. First, Caldwell and Oehlsen (2023) estimates labor supply elasticities using random variation generated by an experiment that allowed drivers to work for one week with 10-50% higher earnings per trip. Because the experiment explicitly highlighted the potential gains from additional driving, it is natural that drivers may have been more responsive in that setting. Second, our sample of drivers tends to have a stronger attachment to work (i.e., they consistently drive more hours per week than the average Uber driver), likely making them less responsive to changes in market hourly pay.

Our data also allow us to analyze whether the quality of the service that drivers offer is affected by college activities. To this end, we performed a similar descriptive event-time analysis as in Section 3.1 but with the ratings the drivers received from their passengers and the share of income from tips as dependent variables. Appendix Figure H.3 shows no negative impact on ratings, while Appendix Figure H.4 shows a small drop in the fraction of earnings coming from tips, but the economic impact is extremely small. Thus, we conclude that college activities have no meaningful impact on the quality of service.

Overall, our findings indicate that the "short-run" opportunity cost of coursework in enrolled weeks is modest: approximately \$180 per month relative to average monthly Uber earnings of about \$1,900 among students taking ASU classes. At the same time, we do not observe any meaningful negative impact of college activities on the quality of work performed. Extrapolating these results to all courses required for a bachelor's degree suggests that the average student in the program would forgo roughly \$7,560 in earnings to complete the degree.[31] In Section 9.2, we combine this opportunity-cost estimate with evidence on longer-run credit accumulation and graduation rates to assess the wage return required for participation in the program to be financially worthwhile.

Table 6: Determinants of study effort

	(1) Study Hours in c	(2) Study Hours in c	(3) Placebo Test
Avg. study hours in c, peers	1.134***	-	1.022***
	(0.027)		(0.018)
Sum of avg. study hours in $-c$, peers	-0.009	-	-0.007***
	(0.006)		(0.002)
Assignments graded in c	-	0.046***	-
		(0.015)	
Sum of assignments graded in $-c$	-	-0.012**	-
		(0.005)	
Market level active hourly pay, currently elig.	-0.008**	-0.029***	-0.001
	(0.004)	(0.004)	(0.001)
Observations	55293	55293	555752
First-stage F-statistic	-	586.82	-
Individual Fixed Effects	✓	✓	✓
Classroom Fixed Effects	✓	✓	✓
Month-Year Fixed effects	✓	✓	✓
Weather controls	✓	✓	✓
Mean dep. var.	1.58	1.58	-
Avg. study hours in c, peers	1.28	-	-
Sum avg. study hours in $-c$, peers	1.55	-	-
Assignments graded in c	-	2.21	-
Sum assignments graded in $-c$	-	2.78	-
Market level active hourly pay, peers	33.56	33.56	-
Elasticity w.r.t. peer coursework	0.919	0.064	
Elasticity w.r.t. peer earnings	-0.173	-0.626	

Notes: The dependent variable in each column is weekly study hours in course c. "Avg. study hours in c, peers" is a proxy for coursework that denotes the average Canvas hours of non-Uber-students in course c. The variable "Sum of avg. study hours in $-c$, peers" denotes the sum of the average hours per week that peers spent on Canvas across all other courses besides c. "Assignments graded in c" and "Sum of assignments graded in $-c$" denote the number of assignments graded in course c, and all other courses $-c$ for a student in a given week. The results in Column (2) are from a 2SLS specification where "Assignments graded in c" and " Sum of assignments graded in $-c$" are instrumented with the average graded assignments for non-Uber peers in c and $-c$, respectively. "Market level active hourly pay, currently elig." denotes the average active hourly pay in a given week of drivers who are currently eligible for the ASU-Uber partnership, but not enrolled. Specifications include multiple sets of fixed effects. Weather controls include second-order polynomials in total weekly rainfall and snowfall. Each regression model also includes a flexible control for the number of days in the week for the course. Elasticities are calculated by multiplying the relevant coefficient estimate by the ratio of mean peer coursework or market level hourly earnings to the mean of the dependent variable. Standard errors are clustered at the classroom level. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

6.2 Labor Market Conditions and Study Effort

Table 6 presents results from estimating the effects of labor market conditions and college course demand on Uber-students' Canvas hours. Evidence on how variation in weekly earn-

[31] We report this figure as a mechanical projection of the short-run, intensive-margin opportunity cost over a degree's worth of courses; it is not an estimate of the cumulative effect of enrollment on labor supply. We calculate the total opportunity cost of receiving a degree by assuming that students need to complete 40 courses to graduate (a minimum of 120 credits are required and over 85 percent of the courses taken in our sample are worth 3 credits). Multiplying 40 courses by the product of average number of peer study hours per course (11.47 hours) and our estimate of work hours forgone per peer Canvas hour (-0.68) yields a total reduction of roughly 312 work hours, which we then scale by average hourly earnings while taking courses ($24.23).

ings impacts study effort is scarce. Thus, the estimates in Table 6 are unique in that they quantify the extent to which Uber-students' study time responds to short-run changes in earnings opportunities. Column (1) shows that study effort is only modestly responsive to transitory changes in labor market conditions. A \$1 increase in average hourly pay decreases study time by less than a minute per course-week. Equivalently, a 10% increase in average hourly pay reduces study time for the average Uber-student by about 1.7% per course. This finding suggests that study time is not highly sensitive to changes in labor market conditions.[32] However, this result could be specific to our context, given that flexibility may allow drivers to spend more time on the Uber platform when earnings are high without meaningfully decreasing their study time.

Column (1) also shows that average peer Canvas hours have a nearly one-to-one relationship with Uber-student study hours, suggesting that our measure of course demand effectively captures the variation in college work across weeks.[33] By contrast, we find little evidence of crowd-out across courses, given that an increase in coursework demand in other courses does not meaningfully reduce study time in course c. Column (2) presents a related specification that uses the number of assignments graded as an alternative measure of course demand. The results imply that an additional assignment in course c increases weekly average study hours in course c by about 2.8 minutes (0.046 hours), or by around 3% relative to the mean.

Column (3) presents a placebo test designed to assess whether market-level hourly pay is proxying for unobserved shocks that also affect online study behavior. For example, suppose weather conditions (which we control for) simultaneously affect study hours and earnings. In that case, we should observe a mechanical correlation between Uber's market hourly pay and study hours for students who do not drive with Uber. To this end, we repeat the specification in Column (1) but on the sample of students attending ASU Online classes

[32] A potential concern for external validity is that Uber students may be positively selected for their time management, making our results limited to our sample. However, Appendix C.2 shows no evidence of smoother within-semester study patterns among Uber-students relative to their classmates; if anything, their weekly Canvas hours are more variable within the same course offering. This suggests that our findings are unlikely to be driven by a subgroup with unusually strong time-management skills.

[33] The bottom subfigures of Appendix Figure H.2 present the distribution of course effort elasticity with respect to coursework and earnings, calculated based on the average coursework and earnings of each Uber-student.

but not participating in the ASU-Uber program. Reassuringly, we find a coefficient of zero for these students.[34] Additionally, Appendix Table G.9 replicates the analysis in Table 6 using weekly clicks and logins as alternative measures of study effort and yields qualitatively similar results, indicating that our findings are not sensitive to the choice of effort measure.

Taken together, Tables 5 and 6 quantify the short-run trade-offs that Uber-students face in terms of work-study time allocation in this flexible environment. Because our identifying variation is week-to-week, both the elasticity and level estimates should be interpreted as within-semester, intensive-margin allocation effects. They capture how current work changes when coursework becomes more demanding and how current study effort changes when labor market opportunities improve. In this sense, the estimates are informative about short-run strategic scheduling and marginal substitution, but they should not be interpreted as the cumulative year-level crowd-out of work and schooling activities.[35]

6.3 Heterogeneity

In Appendix D, we explore the extent to which short-run substitution estimates are heterogeneous across demographics, field of study, driving behavior, and financial need. Broadly, we find limited differences across observable characteristics, with the notable exceptions that very active drivers (i.e., individuals who drive, on average, 25 or more hours per week when not enrolled in ASU courses) and STEM majors reduce working hours to a greater degree in response to coursework demands, indicating that the short-run opportunity cost of taking coursework is somewhat higher for these subgroups.

[34] As an additional robustness check, we re-estimate Table 6 specification with week-year fixed effects. This is feasible because identification can still come from differences across geographic markets. However, because our measure of market-level active hourly pay is constructed at the city-week level, these more saturated time controls absorb most of the common time-series variation in that regressor. Consistent with this mechanical implication, the coefficient on market-level hourly pay becomes close to zero and statistically insignificant, while the estimated effects of course demand on study time remain very similar to our baseline results (Appendix Table G.8). We therefore retain the month-year fixed-effects specification in Table 6 as our baseline, and view the week-year fixed-effects results as further evidence that Uber-students' study effort is not meaningfully driven by short-run labor market shocks.

[35] While our main estimates should be interpreted as short-run, intensive-margin responses, Appendix C.1 provides complementary descriptive comparisons of weekly Uber outcomes during enrolled and non-enrolled weeks. These comparisons do not provide causal estimates of the extensive-margin effect of course enrollment on work, but they provide a useful point of comparison for our short-run estimates by showing how Uber outcomes differ in weeks when drivers are enrolled in ASU courses relative to weeks when they are not. Reassuringly, these descriptive patterns are qualitatively consistent with the main results and are of the same general magnitude, although the enrolled-versus-non-enrolled differences are larger.

6.4 Dynamic Responses and Anticipation

One potential concern with our contemporaneous weekly estimates is that coursework demands may be partly predictable over the academic term. If students anticipate upcoming spikes in coursework—for example, around midterms or final exams—they may adjust their work schedules in advance, which may lead us to understate the total labor-supply response to coursework demands in our baseline estimates.

To assess this possibility, we augment our baseline weekly labor-supply specifications with one-week leads of our course-demand proxy and market-level hourly pay to examine whether future workload predicts current work behavior. Appendix Table G.10 reports the results. If students systematically reduce labor supply in advance of predictable increases in coursework, future course demand should negatively predict current driving hours. Instead, we find no evidence of anticipatory adjustments. We estimate a small and statistically insignificant relationship between next week's course demand and current driving hours. We similarly find no evidence that students adjust current study hours when coursework demands or wages are higher in the following week (see Appendix Table G.11). These findings suggest that our baseline estimates are driven by contemporaneous within-week reallocation and scheduling responses rather than systematic one-week-ahead anticipation.

The lag specifications in Appendix Tables G.10 and G.11 do, however, show some evidence of adjustment across adjacent weeks. Higher market pay in the previous week yields somewhat lower current work and study effort, consistent with partial intertemporal reallocation following favorable labor-market conditions. Lagged course demands also lead to a small increase in current study hours, suggesting modest persistence in academic effort. Importantly, accounting for these lagged responses does not meaningfully change the overall magnitude or interpretation of our baseline results.[36] Overall, the results suggest that adjustment occurs largely through contemporaneous within-week reallocation, with limited spillovers across adjacent weeks.

[36] Across the lead and lag specifications, the estimated effects of coursework demands remain broadly consistent with the baseline results in Tables 5 and 6. For market-level hourly pay, including the lag shifts part of the estimated response into the subsequent week. Although the contemporaneous coefficients differ from the baseline estimates, the sums of the current and lagged coefficients are nearly identical to them.

7 The Role of Flexibility

This section assesses whether Uber-students actively adjust the timing of their work when course demands increase, and whether they value this flexibility.

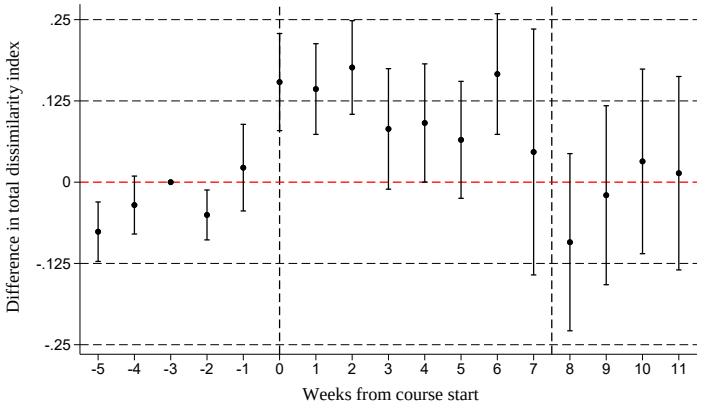
7.1 How Do Working Hours Adjust in Response to Coursework?

To investigate whether Uber-students actively re-time their work schedules when course demands change, we first construct a dissimilarity index that captures deviations in the driving schedules of Uber-students from their typical driving patterns when not enrolled in classes. In particular, our data allow us to observe how drivers distribute their time on the Uber platform across 28 six-hour periods in each week; for example, the first period is Monday morning from 12:00 AM to 6:00 AM and the second is Monday from 6:00 AM to 12:00 PM. We use these data to construct an index that reflects overall deviations from a driver's usual driving schedule when not enrolled at ASU. In a given week, t, Uber-student i's total dissimilarity index is equal to:

$$D_{i,t} = \frac{1}{2} \sum_{p=1}^{28} \left| \frac{\bar{m}_{i,p}}{\bar{M}_i} - \frac{m_{i,p,t}}{M_{i,t}} \right|, \tag{10}$$

where p refers to one of the 28 six-hour periods, $m_{i,p,t}$ is the total minutes driven in period p in week t, $M_{i,t}$ is the total number of minutes driven in week t, $\bar{m_{i,p}}$ is the average minutes driven across all p periods in non-ASU weeks (i.e., when not enrolled in classes), and $\bar{M}_i$ is the average number of minutes driven across non-ASU weeks. This measure compares the within-week allocation of driving time in week t to the driver's typical non-ASU driving schedule. Because the index is based on the share of weekly driving minutes allocated to each of the 28 periods, it intentionally differences out proportional changes in total weekly hours. Thus, if a driver simply reduces driving uniformly across the week while preserving the same relative timing pattern, the index does not change. In that sense, the index isolates within-week re-timing of work rather than reductions in total driving hours. Interpreted as a total-variation distance from the driver's typical non-ASU schedule, larger values indicate more active reshuffling of driving time across days and time periods. For example, $D_{i,t} = 0$ implies no change relative to the typical non-ASU schedule, whereas a value of 1 implies a

Figure 3: Variation in within-week driving flexibility during eight-week courses



Notes: OLS coefficient estimates from a version of Equation (1) where the dependent variable is the standardized dissimilarity index and 95% confidence intervals are plotted. The estimating sample excludes individuals who appear to have dropped out of the course. Standard errors are clustered at the classroom level.

fully dissimilar schedule. For ease of interpretation, we standardize the index in the empirical analysis.

Figure 3 presents results from a specification analogous to the event-time exercises previously presented but using the (standardized) dissimilarity index described above as the dependent variable. Because the index is standardized, each coefficient can be interpreted in standard-deviation units relative to the omitted event week (Week -3). Positive coefficients indicate that students' driving schedules deviate more from their typical non-enrollment schedules. The estimates show that driving schedules become more dissimilar once courses begin. In the first week of an eight-week course, the dissimilarity index increases by roughly 14% of a standard deviation. The increase is largest near the beginning and end of the course, when Canvas engagement also tends to be highest, and smaller during the middle of the course. These patterns suggest that students take advantage of within-week scheduling flexibility to adjust the timing of work when course demands are highest.

The magnitude of this change is economically meaningful. A simulation exercise detailed in Appendix E shows that the average course-week increase in the dissimilarity index is comparable to the change generated by randomly perturbing a non-course week's driving schedule, reassigning 40% of weekly driving time to two randomly selected six-hour windows. We therefore interpret the results as evidence that students actively re-time work during

enrolled weeks, rather than simply reducing hours uniformly across the week.[37]

Appendix C.2 provides complementary evidence that Uber-students also make use of flexibility in their coursework. Relative to other ASU Online students in the same course offerings, Uber-students exhibit greater week-to-week variability in Canvas hours. Taken together with the results above, the patterns are consistent with Uber-students using flexibility on both margins by re-timing work within the week and allowing study effort to vary across weeks.

7.2 Assessing Students' Valuation of Flexible Schedules

Next, we examine students' valuation of flexibility in college and work schedules. To this end, our survey elicited students' stated likelihood of enrolling in a bachelor's program under eight scenarios that varied along three dimensions: whether the work schedule was flexible or fixed, whether courses were offered online or in person, and whether annual tuition was \$0 or \$6,000.[38]

Table 7 reports average stated enrollment probabilities separately for Uber-students and their class peers. Uber-students, often from less privileged backgrounds (see Table 1), showed a greater sensitivity to education costs than their classmates, with enrollment probabilities decreasing by about 50% when costs rose from \$0 to \$6,000, irrespective of the learning and working formats. Students also report a higher likelihood of enrollment in scenarios that combine online learning with flexible work schedules. For instance, moving from the fully rigid environment—fixed work schedules and in-person classes—to the fully flexible environment—flexible work schedules and online classes—substantially increases the reported likelihood of enrollment. Therefore, these findings strongly suggest that this sample of students places considerable value on flexible formats when considering college enrollment.

The table also suggests that flexibility in learning modality is especially important. For

[37]These event-time estimates translate to an average treatment effect across weeks that can be summarized as follows: when the average study time of their peers increases by one hour per week, on average, drivers deviate from their typical schedule by roughly 4.1% of a standard deviation. Moreover, Appendix Figure H.5 presents similar results where we construct the dissimilarity index specifically for typical peak hours periods (i.e., Friday and Saturday afternoons and nights). The results from the figure suggest that these driving behaviors during those hours are particularly likely to adjust during course weeks.

[38]Respondents were asked to report their percent chance of enrolling under each combination of a fixed or flexible work schedule and online or in-person coursework, holding work hours and earnings constant. They answered all four questions first under an annual education cost of \$0 and then under a cost of \$6,000. The latter is approximately the regular annual cost faced by an ASU working student.

Table 7: Average Stated Likelihood of Enrollment by Cost and Flexibility Scenario

	(1) Uber-students	(2) Class Peers	(3) Diff. P-values
Enrollment likelihood with cost = $0			
Fixed work schedule x In-person Classes	30.51	32.80	0.40
Fixed work schedule x Online classes	73.12	84.92	0.00
Flexible work schedule x In-person classes	47.62	55.38	0.01
Flexible work schedule x Online classes	88.95	94.22	0.00
Enrollment likelihood with cost = $6,000			
Fixed work schedule x In-person classes	17.50	26.10	0.00
Fixed work schedule x Online classes	33.18	70.13	0.00
Flexible work schedule x In-person classes	25.51	42.82	0.00
Flexible work schedule x Online classes	43.74	78.82	0.00
Observations	243	344	

Notes: "Uber-students" denotes ASU students who participate in the ASU-Uber partnership. "Class peers" denotes students enrolled in the same classes as Uber-students. "Diff. P-values" reports p-values for differences between columns (1) and (2).

both Uber-students and their class peers, the mean likelihood of enrollment increases by roughly a factor of two or greater going from in-person classes to online classes (when keeping the work schedule fixed).

Appendix F uses these responses to estimate model-based willingness to pay (WTP), following the expected-utility framework of Aucejo et al. (2023). Estimated WTP for flexible learning is two to three times larger than WTP for a flexible work schedule. Because Uber-students are more sensitive to education costs, however, their estimated WTP for both forms of flexibility is lower than that of their class peers.

8 Effort Measures and Academic Performance

A natural question is how work and study effort affect students' weekly and overall course-level performance. Panel A of Table 8 presents results from estimating different versions of equation (8), which focuses on course weekly performance, as measured by the share of points earned in the week (i.e., $\frac{\text{points earned}}{\text{points possible}} \times 100$). Although this normalization removes the mechanical relationship between points at stake and points earned, common course-week difficulty may still affect both peer study time and Uber-students' normalized performance; we therefore interpret the estimates presented in Panel A as supportive evidence on the performance consequences of study-time adjustments. Column (1) considers total Uber (work) hours, while Column (2) refers to a specification that accounts for Canvas (study) hours. Both columns present results from specifications that instrument Uber-students'

weekly hours on Uber and Canvas with the average market-level pay for eligible drivers and average hours on Canvas of class peers (excluding other drivers in the ASU-Uber program), respectively. The results in Panel A suggest that spending more hours on the Uber platform may have a small negative (but economically small) effect on weekly course performance: each additional hour of driving is associated with a (small) decline in the share of points earned of about 0.6 points on a 0-100 scale. As expected, Column (2) shows that an additional hour on Canvas increases the point share attained. Though seemingly modest, we later show that this effect appears large enough to meaningfully affect aggregate course performance.

Next, we explore how aggregate effort throughout the term translates to overall course performance. Panel B of Table 8 shows results corresponding to specifications that include classroom (e.g., a specific section of ECN 101) and individual-by-term fixed effects, as described in Section 5.3. As with Panel A of the table, Column (1) considers total Uber hours, and Column (2) presents results from a specification that accounts for Canvas hours. In Column (1), we instrument an individual's total driving hours throughout the course with the average market-level hourly pay for drivers who were eligible for the ASU-Uber program but did not enroll, averaged across all course weeks. The specification presented in Column (2) instruments a student's Canvas hours with average total hours on Canvas of class peers. For robustness, Appendix Tables G.13 and G.14 present results of similar specifications that examine how work and study time affect course completion and pass rates.

Three main takeaways emerge from the results in Panel B. First, we find no evidence that additional driving hours reduce course grades. Second, based on the estimates from the IV specifications in Panel B, we find that a 10% increase in the average total Canvas hours (i.e., 1.35 hours) increases course grades by 0.11 points (i.e., 1.35×0.078), approximately one-tenth of a grade point, which is consistent with the positive effects found in Column (2) of Panel A. This result indicates that Canvas hours are strongly associated with course performance, suggesting that, though imperfect, our measure of study hours is highly relevant. Finally, using the estimates in Panel B, it is possible to provide a back-of-the-envelope calculation to quantify how changes in labor market conditions (i.e., market hourly pay) impact students' academic performance through a reduction in study hours. Combining the estimates from Tables 6 and 8, we find that a 10% increase in average market-level hourly pay will lead to

Table 8: The effect of work and study effort on course outcomes

Panel A: Weekly performance		
	(1) Point Share	(2) Point Share
Work hours	-0.614**	-
	(0.293)	
Study hours	-	1.020***
		(0.369)
Obs.	55293	55293
First-stage F-stat.	39.57	2058.84
Individual Fixed Effects	✓	✓
Classroom Fixed Effects	✓	✓
Month-Year Fixed Effects	✓	✓
Weather Controls	✓	✓
Avg. dep. var.	86.38	
Avg. study hours	1.58	
Avg. work hours	17.71	
Panel B: Final course grades		
	(1) Course Grade Points	(2) Course Grade Points
Total work hours	0.001	-
	(0.003)	
Total study hours	-	0.078***
		(0.016)
Obs.	4259	4259
First-stage F-statistic	23.80	73.61
Adjusted R^2	0.61	0.64
Classroom Fixed Effects	✓	✓
Student-by-Term Fixed Effects	✓	✓
Avg. dep. var.	2.54	
Avg. total hours on Canvas	13.50	
Avg. total Uber hours	142.16	

Notes: Panel A—The dependent variable, "Point share", is on a scale of 0 to 100. "Work hours" refers to the number of hours spent on the Uber platform in a given week, and "Study hours" denotes the number of hours a student was connected to Canvas for a given class each week. Each regression model includes individual, classroom (e.g., a specific section of ECN 101), and month-year fixed effects. The model in Column (1) is estimated via 2SLS where "Work hours" is instrumented with the weekly average market-level pay for Uber drivers who were eligible for the ASU-Uber program but did not enroll. The model in Column (2) is estimated via 2SLS where "Study hours" is instrumented with the average Canvas hours for non-Uber peers in the course for a given week. The mean point share differs from the mean reported in Table 2 because weeks with no possible points are coded as receiving the maximum point share. The results are robust to excluding these weeks or, alternatively, coding them as receiving the minimum point share. Panel B—The variable "Total work hours" denotes the total hours an Uber-student spent connected to the Uber platform throughout the course. "Total study hours" refers to the sum of individual Uber-student's own Canvas hours throughout a given course. Each regression model includes classroom, and individual-by-term fixed-effects. The sample excludes all terms that were meaningfully disrupted by the COVID-19 pandemic. The model in Column (1) is estimated via 2SLS where "Total work hours" is instrumented with the average (across weeks) of average market-level pay for Uber drivers who were eligible for the ASU-Uber program but did not enroll. The model in Column (2) is estimated via 2SLS where "Total study hours" is instrumented with the sum (across weeks) of average Canvas hours for non-Uber peers in the course. All standard errors are clustered at the classroom level. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

a small decrease in course grade performance, roughly 0.016 grade points.[39]

[39]The point estimates from Table 6 imply that a 10% ($3.36) increase in average market-level hourly pay of peers would lead to a reduction in weekly study hours by 0.027 hours (i.e., $3.36×-0.008). If we multiply this number by 7.5 (most courses are either 7 or 8 weeks long), then the decrease in the total number of study hours for a given course is around 0.202 hours. Finally, if we use the estimate from Column 2 of Table 8 (Panel B), jointly with 0.202, we obtain the 0.016-point decrease in the course grade.

Overall, the patterns in Panels A and B suggest that flexible work and study arrangements may allow Uber-students to adjust their schedules in ways that largely preserve course performance.

9 Perceived Returns and Degree Progress

9.1 Students' Perceived Returns to Program Participation

Our survey collected students' perceptions about their expected graduation rates and future income under various scenarios, allowing us to characterize the ex-ante value students attribute to the program. These subjective beliefs—whether or not they are ultimately realized—are central for understanding participants' decisions to enroll in the program.

Appendix Table G.15 reveals that students anticipate an average graduation rate exceeding 90%. We also inquired about the expected graduation rate of transfer students had they remained at their previous institutions. Students' responses indicate a marked improvement in anticipated graduation rates due to transferring, with an increase of about 25 percentage points.[40] Whether these expectations are ultimately realized is a separate question, but ex-ante these students are quite serious about completing their programs. Finally, the bottom panel of Table G.15 presents the ratio of the students' projected annual income in six years, conditional on obtaining a bachelor's degree or not (i.e., dropping out of the program). On average, students foresee a 90% return from earning a bachelor's degree, highlighting the significant perceived value they attach to completing the program.

It is admittedly too early to determine whether students will ultimately realize such substantial returns to program participation; in addition, we cannot track participants' longer-run outcomes individually beyond May 2021. However, it is crucial to note that the online programs examined in this paper differ markedly from many conventional short online post-secondary offerings, as we examine bachelor's degree programs offered by a large public university that does not distinguish between online and in-person degrees.[41] Moreover, re-

[40] The conditional means underlying this transfer-student comparison are not reported in Appendix Table G.15.

[41] Appendix Table G.16 shows that 94% of Uber-students considered ASU education to be either better (48%) or comparable (46%) to their prior institutions' quality, with similar perceptions also reported by their class peers. While these shares should be interpreted with caution because they are conditional on students currently enrolled at ASU Online, they suggest that Uber-students are not trading off college quality for more flexible, free education.

cent employer surveys further suggest growing recognition of online degrees as equivalent to traditional, in-person qualifications.[42] These developments suggest that online degrees are increasingly viewed as credible credentials, which helps contextualize why students in our sample report substantial expected returns to completing an ASU Online degree.

9.2 Graduation Rates and Credit Accumulation

The survey evidence above captures participants' perceived returns to degree completion and their expectations about educational progress. We next complement these subjective measures with administrative evidence on credit accumulation and graduation. Although data limitations prevent us from tracking these outcomes for the drivers in our analytic sample beyond May 2021, we also obtained aggregate graduation and credit-accumulation statistics for the first four cohorts of ASU-Uber program participants, which include all three entry cohorts represented in our high-frequency sample.

Appendix Table G.17 provides a summary of both persistence and degree progress among the first four ASU-Uber entry cohorts. As of December 2025, 27.2% of participants in the first four cohorts had graduated. Among the first two cohorts, for which five-year follow-up is complete, the cohort-size-weighted five-year graduation rate was 27.9%, compared with a graduation rate of 30.2% as of December 2025. The difference between these rates suggests that a meaningful share of participants graduate more than five years after entering the program, which is unsurprising given the program's flexibility and the competing demands faced by working students.[43] At the same time, the average entrant accumulated 81 ASU credits, or roughly two-thirds of the 120 credits required for a bachelor's degree. These figures indicate that, although many participants do not complete the program within five

[42]For instance, the National Association of Colleges and Employers found that 87.4% of employers who track degree modalities have hired graduates with online degrees, offering them the same starting salaries as those awarded to candidates with traditional degrees.

It is also important to mention that the returns to a college degree remain high even for marginal students (Zimmerman, 2014). While Deming et al. (2016) find that degrees from online for-profit institutions lead to lower employer callback rates than comparable degrees from nonselective public schools, our context differs notably in two respects. First, ASU is a widely recognized public institution, well regarded for its online programs. Second, ASU transcripts do not specify whether degrees were earned online or in-person, potentially mitigating modality-related factors.

[43]As a benchmark, the six-year bachelor's degree graduation rate for full-time, first-time bachelor's degree-seeking undergraduate students at ASU Digital Immersion is 24% (National Center for Education Statistics, 2023). This comparison suggests that Uber participants enrolled at ASU may have graduation rates somewhat higher than the relevant online undergraduate benchmark, although that population and follow-up window differ from those in our data.

years, the average entrant nevertheless makes substantial progress toward a degree, which is important for interpreting both the costs and potential benefits of participation.

Relying on this additional information on credit accumulation and graduation rates, we can calculate how long it would take the average entrant to offset the opportunity cost of participation through higher post-graduation earnings. Scaling our estimated full-degree opportunity cost of $7,560 in proportion to 81 accumulated credits implies expected forgone earnings of roughly $5,100 per entrant. If we conservatively assume that earnings gains accrue only to participants who graduate, then expected benefits must likewise be scaled by the observed graduation probability of 27.2%. Assuming an illustrative 15% post-graduation wage gain—which falls close to the middle of estimates in the literature on the returns to college *enrollment* (Mountjoy, 2026; Zimmerman, 2014)—and annual Uber earnings of approximately $24,000, the average entrant would offset expected opportunity costs after roughly five years. This payback calculation provides a simple entrant-level benchmark for evaluating whether the program's longer-run earnings gains are large enough to justify its short-run opportunity costs.[44]

10 Conclusion

This is the first study to quantify how college activities and labor market conditions affect labor supply and study effort at a high frequency. Our findings show that an additional hour of coursework displaces roughly two-thirds of an hour spent on the Uber platform (about 1.3 fewer completed trips) or, equivalently, that a 10% increase in weekly coursework reduces platform hours by about 1%. However, at observed levels of coursework and labor supply, the implied forgone Uber earnings are modest, with average coursework during enrolled weeks reducing earnings by about $41 per week, or less than 10% of weekly Uber earnings. Students also adjust their driving timing as course demands increase, and we find little evidence that coursework reduces the quality of their driving. Conversely, study effort remains mostly unaffected by short-run fluctuations in labor market conditions: a 10% increase in average hourly pay reduces study time for the average Uber-student by about 1.7% per course.

[44] We present this as an undiscounted payback benchmark rather than a formal present-value calculation because discounting would require additional assumptions about the timing of both costs and benefits. In particular, students accumulate credits over different horizons, and post-graduation earnings gains may begin at different points depending on time to degree.

Finally, we find no detectable reduction in final course grades associated with additional work hours.

Survey evidence further indicates that students in our sample place substantial value on flexibility in learning environments and report meaningful expected returns to completing an online degree. We interpret these patterns as suggesting that flexibility may matter for participation in higher education, while recognizing that our data do not allow us to identify the program's causal effect on college demand or to conduct a full welfare analysis.

Data Availability

Code replicating the tables and figures in this article can be found in Aucejo, Perry and Zafar (2026) in the Harvard Dataverse, `https://doi.org/10.7910/DVN/XXXXXX`.

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Supplementary Material for "Balancing College and Work Activities: The Gig Economy Meets Online Education"

By Esteban Aucejo, Spencer Perry, and Basit Zafar

A Additional Data Description

A.1 Select Variable Definitions

In this subsection, we provide an overview of key variables used in our analysis. In terms of relevant measures of study effort, we directly observe the amount of time that students are connected to a course's Canvas page, as well as the number of logins and clicks that a student generates each week. To construct the measures of course demand in a given week, we calculate the average amount of time spent on Canvas by all course peers (i.e., non-Uber program participants).

For work effort/labor supply measures, we observe how much time a driver is connected to the Uber application and how many weekly trips they complete. Note that, at the driver level, we cannot distinguish between active and inactive time, where active time refers to time spent picking up and completing rides, and inactive time includes waiting for ride requests (while connected to the application). At the market level, we observe average active and inactive time connected to the Uber application for all drivers who were eligible for the ASU program but did not enroll. We also observe average weekly earnings for this same group. For our primary analysis, we calculate average market-level earnings (often denoted as "market level active hourly pay, currently eligible") as the ratio of average weekly earnings to average active hours for those eligible drivers who did not enroll in the program. For robustness, we also re-conduct our analysis using a measure of market-level earnings that is equal to the ratio of average earnings to average total hours (i.e., the sum of active and inactive hours). The results of the paper are both qualitatively and quantitatively consistent, regardless of the definition of market-level pay. We prefer active hourly pay because we think this is the more salient measure of compensation for drivers. There is substantial variation in market-level weekly hourly pay, where the standard deviation of this variable is \$5.5 in the sample. In addition, there is considerable variation across cities, as well as within cities across weeks.

The standard deviation in weekly hourly pay across cities is $4.3, and within cities across weeks is $3.5.

A.2 Sample Restrictions

The primary way in which we restrict our data to construct our estimation sample is to limit the weeks included in our analysis. In particular, we drop all weeks between March 16, 2020, and January 4, 2021. The motivation for doing this is that the COVID-19 pandemic and related stay-at-home orders led to a massive disruption in Uber driver activity. We include the Spring term of 2021 in our analysis because, by that point, Uber driving behaviors roughly returned to pre-pandemic levels, and a large share of our sample comes from that term. In addition to dropping these observations, we also exclude "Dynamic" courses from our estimating sample. These courses are often only one week long and make up a very small share of our Uber-student sample. Besides these restrictions, we do not place any other explicit filters on our estimating sample.

For our estimation of Equation (1) (i.e., analysis of data variation), we also make additional sample restrictions. Specifically, we exclude individuals who appear to have "dropped out" during the course. We determine likely dropouts based on their final course grades (e.g., incomplete or withdrawals) and their activity. Specifically, if a student receives a failing grade, an incomplete, or a withdrawal *and* has a spell of inactivity for at least the final three weeks of the course, we consider them a likely dropout. Though excluding these observations from our main results has little effect, they do matter in the estimation of our data variation analysis since dropouts meaningfully attenuate our estimates in weeks after they drop out (since their driving hours are relatively high and study hours relatively low due to dropping out).[1]

B Market-Level Pay and Hotel Occupancy

To understand what drives the variation in market-level pay for our main results, we collected additional data on hotel occupancy for a subset of the Uber markets. The data come from STR Inc., a hotel analytics company founded in 1985 as Smith Travel Research, the recognized leader in hotel industry analytics and provides market data, including supply,

[1] Appendix Tables G.18 and G.19 present results where we re-estimate our main specifications but limit the sample by excluding likely dropouts.

demand, and market share information on a global scale. For our purposes, STR provides comprehensive daily hotel occupancy data from 2018 to 2023 across 95 Metro Markets. Our analysis's primary variables of interest are daily occupancy rates for each market. To convert the daily occupancy data to a weekly frequency, we take the average occupancy rate from Monday through Sunday of each week. However, the results are robust to alternative constructions. We can merge data for 69 of the 111 unique Uber markets, which corresponds to roughly 94% of the observations included in the results for Table 6. With these merged data, we estimate alternative specifications of the following regression:

$$MarketPay_{lt} = \beta_0 + \beta_1 Occupancy_{lt} + \beta_2 Weather_{lt} + \psi_l + \rho_m + \varepsilon_{lt}, \tag{B.1}$$

where l is the labor market, t is the week, the occupancy variable is the mean from Monday through Sunday of the given week, and ψ_l and ρ_m are city and month-year fixed effects. Results reported in Table B.1 show that the market-level weekly pay variable increases as the occupancy rate increases despite including a robust set of controls. To ease interpretation, note that a one standard deviation increase in the occupancy rate (17.5) leads to a 0.25 standard deviation increase in the market-level pay ($17.5 \times .073 = 1.28$ and $1.28/5 = .25$). Overall, these findings suggest that the variation in market pay we use for our analysis responds to shocks to demand.

Table B.1: Relationship between Market-Level Pay and Hotel Occupancy Rate

	(1)	(2)	(3)
Hotel Occupancy Rate$_{lt}$	0.067***	0.074***	0.073***
	(0.003)	(0.003)	(0.003)
Obs.	7845	7844	7609
F-stat.	440.45	526.62	97.57
Fixed-effects	No	Yes	Yes
Precip. controls	No	No	Yes
Mean dep. var.	30.1	30.1	29.9
Std. dev. dep. var.	5.11	5.11	5
Mean occ. rate	53.53	53.53	53.48
Std. dev. occ. rate	17.62	17.62	17.5

Notes: The dependent variable is market-level pay. The variable "hotel occupancy rate" denotes the weekly mean occupancy rate from Monday through Sunday. The hotel occupancy rate data come from STR Inc., a hotel analytics company founded in 1985 as Smith Travel Research. "Precip. controls" denote controls for weather conditions. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

C Addressing Concerns of External Validity

C.1 Descriptive Comparison of Course-Enrollment and Non-Enrollment Weeks

While our main results should be narrowly interpreted as short-run intensive-margin responses, the following table provides a descriptive comparison of weekly Uber outcomes in weeks when drivers are and are not enrolled in ASU courses. Specifically, Table C.1 reports results from regression models where we regress a series of weekly labor market outcomes (i.e., hours worked, total pay, and hourly pay) on indicator variables for the number of ASU courses a driver is enrolled in, in a given week. In these specifications we pool all weeks of driving behavior, so the comparison value of courses is zero (i.e., weeks when drivers are not enrolled at all). We also include individual and week fixed effects.

These results suggest that in weeks where a student is enrolled in two courses, they work 3.17 hours less and earn roughly $74 less than they do in weeks where they are not enrolled in any courses. For reference, the average number of courses taken during a week while enrolled in any ASU courses is approximately 1.9. Interestingly, we find that, conditional on driving any hours, there is little economic difference in the hourly pay that drivers earn when they are and are not taking courses, possibly suggesting that the course flexibility allows them to focus their driving on periods when earning potential is high. Though not causal, these

Table C.1: Relationship between weekly Uber outcomes and academic course load

	(1) Uber hours	(2) Total pay	(3) Hourly pay
1 course	-2.103***	-44.829***	-0.164
	(0.437)	(11.216)	(0.166)
2 courses	-3.166***	-74.260***	-0.025
	(0.392)	(10.262)	(0.148)
3 courses	-4.212***	-105.270***	-0.236
	(0.607)	(15.665)	(0.225)
4 or more courses	-4.657***	-123.119***	0.156
	(1.018)	(29.763)	(0.491)
Obs.	145982	145982	124043
Mean dep. var.	22.04	479.86	21.76
Std. dev. dep. var.	18.28	441.47	8.44
Adjusted R^2	0.41	0.37	0.50
Individual and week fixed effects	Yes	Yes	Yes

Notes: This table reports estimates from regressions of weekly Uber outcomes on indicators for the number of ASU courses in which an Uber-student is enrolled during the week. The omitted category is weeks in which the driver is not enrolled in any ASU courses. "Uber hours" denotes total hours connected to the Uber platform, including both active and inactive time. "Total pay" is the sum of organic pay, incentive pay, and tips. "Hourly pay" is total pay divided by total Uber hours and is therefore observed only in weeks with positive Uber hours. The "4 or more courses" category combines weeks with four or more enrolled courses. All specifications include individual and week fixed effects. Standard errors, clustered at the driver level, are reported in parentheses. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

estimates provide a useful descriptive benchmark for the weekly opportunity cost of course enrollment. Reassuringly, these estimates are qualitatively consistent with the main results presented in Section 6.

C.2 Uber-Students Do Not Smooth Study Hours more than Non-Uber Peers

A potential concern with our findings is that Uber-students may be positively selected on their ability to smooth coursework over the semester or manage their time more effectively than other online students. If that were the case, the estimates we obtain in this setting might have more limited relevance for the broader population of online learners. To assess this issue, we compare the within-semester pattern of study time for Uber-students and their non-Uber peers.

For each student-course pair, we construct two measures of within-semester dispersion in weekly Canvas hours: the standard deviation and the interquartile range. We then regress each measure on an indicator for being an Uber-student, both without controls and in specifications that include course, term, and session fixed effects, so that comparisons are made within the same course offering. The results presented in Appendix Table C.2 show no evidence that Uber-students are smoothing their study time more effectively across weeks than their classmates. To the contrary, the Uber-student coefficient is positive and statistically significant for both dispersion measures in the fixed-effects specifications, indicating greater week-to-week variability in study time among Uber-students.

This analysis alleviates the concern that our findings are driven by a uniquely selected group of students who are inherently better at smoothing coursework across weeks. Instead, the evidence is more consistent with Uber-students taking greater advantage of the flexibility offered by online courses, possibly due to the variability in the opportunity cost of their time driven by within-week shifts in labor demand.

Table C.2: Within-Semester Variability in Study Time, Uber vs. Non-Uber-students

	Std. Dev. of Weekly Canvas Hours		IQR of Weekly Canvas Hours	
	(1)	(2)	(3)	(4)
Uber student	0.397***	0.274***	0.445***	0.308***
	(0.027)	(0.012)	(0.034)	(0.018)
Observations	365069	364996	391246	391157
Fixed effects	No	Yes	No	Yes
Mean dep. var.	1.00	1.00	1.16	1.16
Std. dev. dep. var.	0.85	0.85	1.03	1.03
Adjusted R^2	0.003	0.470	0.003	0.430

Notes: The unit of observation is a student–course pair. In columns (1) and (2), the dependent variable is the standard deviation of weekly Canvas hours across weeks within a course. In columns (3) and (4), the dependent variable is the interquartile range (IQR) of weekly Canvas hours across weeks within a course. The indicator "Uber student" equals one for students in the ASU–Uber sample and zero for other ASU Online students. Columns (2) and (4) include classroom, term, and session fixed effects, so comparisons are made within the same course offering. Standard errors are reported in parentheses. Positive coefficients indicate greater within-semester variability in weekly study time among Uber-students relative to non-Uber peers. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

D Heterogeneity in Work-Study Trade-Offs

To complement our main results, we also examine whether changes in coursework loads and labor market conditions lead to differential responses based on Uber-students' observable characteristics. In particular, we consider i) full-time or part-time driver status, since time constraints may be more binding for full-time drivers; ii) field of study, since certain majors may be more demanding than others; iii) gender, as prior literature demonstrates that labor supply choices may differ by the gender of the driver (Cook et al., 2021; Caldwell and Oehlsen, 2023); iv) financial need status, since labor supply may be more inelastic to changes in coursework demand for this group; v) incoming academic performance (above- vs. below-median incoming GPA), as certain students may be more responsive to coursework demands; and vi) age (under 35 vs. 35 or older), given that schedule flexibility may differ by age (because of family responsibilities etc.).

Panels A and B of Table D.1 examine heterogeneity in the main work and study effort responses. Panel A re-estimates Equation (6), interacting coursework demand with observable student characteristics. Panel B re-estimates Equation (7), interacting market-level hourly pay with the same characteristics.

Column (1) of Panel A shows that full-time drivers reduce labor supply more strongly in response to coursework demand than part-time drivers. We define full-time drivers as those who average at least 25 hours of driving per week when not enrolled in ASU courses. This pattern is consistent with full-time drivers facing tighter time constraints and having less scope to absorb coursework demands without reducing work hours. An alternative explanation is that program eligibility requirements may be more binding for part-time drivers, making their work hours less responsive. However, as discussed in footnote 7, the average work activity of part-time drivers is substantially above the threshold required to maintain eligibility.

Column (2) of Panel A shows that STEM students also reduce work hours more strongly in response to coursework demand. In particular, STEM students' labor supply is roughly 79% more responsive to changes in college activities than that of non-STEM students (-0.645 vs -0.360). This pattern is consistent with STEM courses placing greater demands on stu-

Table D.1: Heterogeneity in the determinants of work and study hours

	(1) Full-time driver	(2) STEM	(3) Female	(4) High-need	(5) High incoming GPA	(6) Age $\geq$ 35
Panel A: Work hours						
Sum avg. study hours all c, peers	-0.381***	-0.360***	-0.497***	-0.554***	-0.465***	-0.489***
	(0.090)	(0.092)	(0.094)	(0.132)	(0.115)	(0.118)
Sum avg. study hours all c, peers × subgroup	-0.237*	-0.285**	0.166	0.079	0.027	-0.005
	(0.137)	(0.136)	(0.157)	(0.144)	(0.134)	(0.132)
Observations	27404	26636	26690	27388	23902	27404
Panel B: Study hours						
Market level active hourly pay, currently elig.	-0.010**	-0.007	-0.007*	-0.011*	-0.007	-0.010*
	(0.004)	(0.005)	(0.004)	(0.006)	(0.005)	(0.005)
Market level active hourly pay, currently elig. × subgroup	0.006	-0.001	-0.015	0.003	-0.003	0.004
	(0.006)	(0.006)	(0.010)	(0.007)	(0.006)	(0.006)
Observations	55293	53731	53975	55266	48263	55293

Notes: This table only reports the relevant coefficients, though the specifications follow equations (6) and (7) but with the inclusion of the relevant interactions with the different subgroups. "Full-time drivers" are those that average at least 25 hours of driving per week when not enrolled in ASU classes. "High-need" indicates that Uber-students are either "High" or "Very High" need based on their FAFSA records, and "High incoming GPA" indicates that the student's incoming GPA was at least 2.72 (the sample median). Appendix Table G.12 shows all the specification coefficients. For Panel A—the dependent variable in each column is hours working on the Uber platform. "Market level active hourly pay, currently elig." denotes the average active hourly pay in a given week of drivers who are currently eligible for the ASU-Uber partnership but not enrolled. "Sum of avg. study hours across all c, peers" denotes the sum of the average hours per week that peers spent on Canvas across all courses. For Panel B—the dependent variable in each column is weekly study hours in course c. "Avg. study hours in c, peers" is a proxy for coursework that denotes the average Canvas hours of non-Uber-students in course c. The variable "Sum of avg. study hours in $-c$, peers" denotes the sum of the average hours per week that peers spent on Canvas across all other courses besides c. All specifications include multiple sets of fixed effects, second-order polynomials in total weekly rainfall and snowfall. Each regression model in Panel B also includes a flexible control for the number of days in the week for the course. In Panel A, standard errors are clustered at the individual level. In Panel B, standard errors are clustered at the classroom level.*, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

dents' time.[2] Finally, we do not find statistically significant differences in the responsiveness of work hours by gender, financial need, incoming GPA, or age.

[2]Our estimates indicate that the return to an individual hour of time spent on Canvas is lower in terms of course grades for STEM students. Specifically, we find that the return to an hour of time on Canvas is only 70% as high as that of non-STEM students. This suggests that STEM majors need to spend relatively more time on Canvas in response to their course activities. In fact, we find that, on average, Uber-students and non-Uber-students in STEM majors spend over two hours more on Canvas for each course.

Panel B examines heterogeneity in the study-effort response to market-level hourly pay. Overall, we find little systematic heterogeneity in how labor market conditions affect Canvas hours. The interaction terms are generally small and statistically insignificant, suggesting that the modest study-effort response to labor market conditions is broadly similar across observable student groups.

Overall, Table D.1 indicates that short-run work-study trade-offs are broadly similar across students with a range of observable characteristics. The clearest exceptions are full-time drivers and STEM students, for whom the short-run opportunity cost of coursework appears somewhat larger.

E Interpreting the Dissimilarity Index

To help contextualize the dissimilarity index results, we first estimate the following regression equation:

$$DissimilarityIndex_{it} = \alpha_0 + \alpha_1 Class_{it} + \alpha_2 MarketPay_{it} + \alpha_3 Weather_{it} + \tau_i + \delta_m + \epsilon_{it}, \quad \text{(E.1)}$$

where $DissimilarityIndex_{it}$ is the standardized (mean zero, standard deviation one) dissimilarity index for person i in week t, $Class$ is an indicator for being enrolled in an ASU course, and τ_i and δ_m are person and month-year fixed effects. Thus, the coefficient of interest, α_1 reflects the average deviation from a driver's standard driving schedule across all course weeks. We estimate $\hat{\alpha}_1 = 0.26$.

To gauge the magnitude of $\hat{\alpha}_1$, we compare it to a simulated within-week reallocation of driving. In each of 100 draws, we randomly select 30% of non-course driver-weeks and, for each selected week, move a fixed share of that week's driving into two randomly chosen six-hour periods while holding the driver's total weekly driving fixed. Specifically, we draw two distinct periods at random from the 28 six-hour periods that make up the week and reallocate 40% of the week's driving into them—20% into each—drawing the reallocated time proportionally from the driver's existing schedule. To help illustrate, consider a driver that drives approximately 20 hours per week in non-course weeks. For this driver, we would add 4 hours to two different random hour bins (e.g., Monday 12 PM-6 PM and Thursday 6 AM-12 PM), drawn proportionally from the rest of the week, so that the driver still drives

20 hours that week but on a more concentrated within-week schedule. We then recompute the standardized dissimilarity index on the reallocated schedules and estimate the following regression limited to non-course weeks:

$$DissimilarityIndexRand_{it} = \beta_0 + \beta_1 Rand_{it} + \beta_2 MarketPay_{it} + \beta_3 Weather_{it} + \tau_i + \delta_m + \epsilon_{it}, \tag{E.2}$$

where the LHS variable is the randomized index (standardized), and *Rand* is an indicator denoting if the observation was treated by randomization. Averaging across the 100 draws, we estimate $\hat{\beta}_1 = 0.25$—close to the observed course-week coefficient $\hat{\alpha}_1 = 0.26$. As a placebo test, we regress the *unperturbed* index on the same, randomly assigned reallocation indicator and obtain an averaged coefficient of essentially zero, confirming that $\hat{\beta}_1$ reflects the simulated reallocation rather than the random selection of weeks. We therefore read the rise in the dissimilarity index during course weeks as comparable in magnitude to reallocating roughly 40% of a driver's weekly driving.

F Students' Willingness to Pay

We propose a straightforward model of expected utility for enrollment, as outlined in Aucejo et al. (2023), that allows us to derive measures of willingness to pay (WTP).[3] Consider the utility U_{is} that student i derives from enrolling under a specific state or scenario, denoted as s. This utility can be expressed as $U_{is} = u_i(X_s) + \epsilon_{is}$, where $u_i(X_s)$ represents the preferences of individual i for a set of characteristics X_s associated with scenario s. These characteristics might include factors like the mode of class delivery (in-person or online), the nature of work schedules (flexible or fixed), and the cost of education. The term ϵ_{is} captures an additional, scenario-specific component of the student's preference. We define ϵ_{is} as $\epsilon_{is} = \delta_i + \varepsilon_{is}$, where δ_i represents the unobserved, scenario-independent utility component for individual i, and ε_{is} corresponds to an idiosyncratic taste shock. In choice models, it is common to assume that ε_{is} is an independent and identically distributed (i.i.d.) Type I extreme value variable, independent of the preferences $u_i(X_s)$.[4] For estimation purposes, we

[3]For a detailed discussion on the validity of this approach and the commonly imposed identification assumptions, see Aucejo et al. (2023).

[4]In our context, the independence of irrelevant alternatives (IIA) assumption is not a concern as students face only two options: enroll or not enroll. ε_{is} represents resolvable uncertainty, meaning uncertainty due

assume that these errors are independent across individuals and conform to a Type I extreme value distribution. Therefore, the probability of student i choosing to enroll in college under scenario s can be modeled as:

$$ProbEnr_{is} = \frac{exp(u_i(X_s) + \delta_i)}{1 + exp(u_i(X_s) + \delta_i)}. \tag{F.1}$$

If we adopt a linear and separable utility specification, the utility function $u_i(X_s)$ can be parameterized as:

$$u_i(X_s) = \alpha_1 \cdot OnlineClass_s + \alpha_2 \cdot FlexibleWork_s + \alpha_3 \cdot Cost_s, \tag{F.2}$$

where $OnlineClass_s$ and $FlexibleWork_s$ denote indicators for online classes and flexible work schedules in scenario s, while $Cost_s$ refers to the student-specific university fees outlined in the scenario.

We estimate this model using the fractional response approach (Papke and Wooldridge, 1996), which accommodates dependent variables within the 0 to 1 range. This method is particularly effective for handling data extremes (0 or 1 values) and is robust in the presence of censored data. The model's parameters are estimated using a quasi-maximum likelihood approach, with the Bernoulli log-likelihood function defined as:[5]

$$\begin{aligned} l_{is}(\alpha, \delta) = ProbEnr_{is} \log &\left(\frac{\exp(u_i(X_s) + \delta_i)}{1 + \exp(u_i(X_s) + \delta_i)} \right) \\ &+ (1 - ProbEnr_{is}) \log \left(1 - \frac{\exp(u_i(X_s) + \delta_i)}{1 + \exp(u_i(X_s) + \delta_i)} \right). \end{aligned} \tag{F.3}$$

Since the model parameters do not have a direct economic interpretation, we calculate the willingness-to-pay (WTP) for online learning as follows:

to unknown factors in the hypothetical survey scenario, which will become known at the time of actual decision-making.

[5]For fractional data, the Bernoulli quasi-maximum likelihood estimator is efficient among a class of estimators that includes all quasi-maximum likelihood estimators in the linear exponential family and weighted non-linear least squares estimators.

$$WTP_{OnlineClass} = -\frac{\alpha_1}{\alpha_3} \tag{F.4}$$

Similarly, the willingness to pay for flexible work schedules is determined as follows:

$$WTP_{FlexibleWork} = -\frac{\alpha_2}{\alpha_3} \tag{F.5}$$

Table F.1 shows WTP estimates for Uber-students and class peers. Results indicate that Uber-students' value for work flexibility (relative to a fixed schedule) is $2,685, while for online learning (relative to in-person classes) it is $5,893. These estimates suggest a strong willingness to pay for flexibility in learning and working formats, though the WTP for flexible working schedules is lower than for online learning. Finally, the ratio (WTP flexible work/WTP online classes) is higher for Uber-students, which is expected given that they chose to have more flexible work schedules.[6]

Table F.1: WTP Estimates for Flexible Work and Learning Schedules

		(1)		(2)		(3)	
	N	WTP Flexible Work	p-value	WTP Online Classes	p-value	WTP ratio (Work/Study)	p-value
Uber-Students	243	$2,685*** (234)	0.000	$5,893*** (271)	0.000	0.46*** (0.04)	0.088
Class peers	344	$6,933*** (569)		$18,239*** (1,177)		0.38*** (0.02)	

Notes: Willingness-to-pay reported in dollars per year. Standard errors in parentheses derived via delta method. The p-value denotes whether differences in WTP between groups (i.e., Uber-students vs. class peers) are statistically significant. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

[6]Survey responses indicate that Uber-students rate their work flexibility as 6.12 on average (on a scale of flexibility from 1 - not flexible at all, to 7 - extremely flexible), while their class peers rate their work flexibility as 3.85.

G Appendix Tables

Table G.1: Correlations of Canvas activity measures for Uber-students

	Active days	Logins	Clicks	Minutes online
Active days	1.000			
Logins	0.822***	1.000		
Clicks	0.635***	0.751***	1.000	
Minutes online	0.562***	0.707***	0.846***	1.000
Observations		55293		

Notes: Table reports correlation coefficients across each metric of weekly Canvas activity. "Active days" corresponds to the number of days in the week with non-zero Canvas activity, "Logins" refer to the number of distinct times a student logged onto the course Canvas page, "Clicks" reflect the number of distinct clicks that were made on various course page links, and "Minutes online" refer to the number of minutes the student spent on the course Canvas page. Sample is limited to the observations included in the analytic sample used in our main regression specifications presented in Table 6.*, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

Table G.2: Weekly Canvas activity by student group, re-weighted control group

	Uber-Students Active in Uber		Uber-Students Inactive in Uber		Class Peers (Re-weighted)		Col (1) - (3)
	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.	Difference
	(1)		(2)		(3)		(4)
Activity							
Days connected to Canvas per course	3.57	(2.30)	3.75	(2.27)	3.60	(2.24)	-0.04***
Clicks per course	75.97	(89.33)	79.04	(85.57)	69.63	(78.45)	6.34***
Hours online per course	1.57	(2.12)	1.61	(2.10)	1.40	(1.85)	0.16***
Outcomes							
Assignments submitted	1.97	(4.14)	1.99	(2.92)	2.03	(3.95)	-0.06**
Assignment share (%)	90.19	(27.59)	90.55	(26.81)	90.00	(27.72)	0.19
Point share (%)	79.40	(29.31)	80.13	(28.66)	79.62	(29.20)	-0.22
Observations	43527		12201		124909		

Notes: The "Class Peers" control group is re-weighted using coarsened exact matching on observables to be more similar to the Uber-student sample. All statistics reported in the table are at the student-course-week level. "Uber-Students: Active in Uber" denotes weeks in which the student completes at least one trip and is enrolled in ASU classes. "Uber-students: Inactive in Uber" denotes weeks in which the student completes zero trips and is enrolled in ASU classes. "Class peers" correspond to students who are enrolled in the same classes as the Uber-students. Assignment and point shares refer to the total number of assignments submitted and points earned relative to the number of assignments due and points possible, respectively.*, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

Table G.3: Student degree progress by enrollee type

	(1)		(2)		(3)
	Uber-Students		Class Peers		Difference
	Mean	Std. Dev.	Mean	Std. Dev.	Mean
Cumulative GPA	2.63	(1.409)	2.94	(1.091)	-0.32***
Credits to date	87.51	(53.233)	85.61	(46.253)	1.90
Observations	1540		64248		

Notes: "Uber-students" denotes ASU students who participate in the ASU-Uber partnership. "Class peers" corresponds to all other students enrolled in the same classes as the Uber-students. "Cumulative GPA" refers to the student's cumulative grade point average at their most recent observed term and "Credits to date" denotes the total number of academic credits on a student's transcript including transfer credits. The sample is restricted to the main estimating sample of Uber-students and their class peers in the same courses. Means are reported for each variable measured at the most recent term in the sample for each student. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

Table G.4: Course-level summary statistics, re-weighted control group

	(1)		(2)		(3)
	Uber Students		Class Peers (Re-weighted)		Col. (1) - (2)
	Mean	Std. Dev.	Mean	Std. Dev.	Mean
Activity					
Total hours on Canvas	14.63	(12.73)	13.09	(11.69)	1.55***
Assignments submitted	18.38	(16.89)	19.03	(17.58)	-0.66**
Outcomes					
Passed course	0.76	(0.43)	0.78	(0.41)	-0.02***
Numeric course grade	2.49	(1.66)	2.57	(1.61)	-0.08***
Observations	6591		14446		

Notes: "Uber-Students" denotes ASU students who participate in the ASU-Uber partnership. "Class peers" corresponds to all other students that are enrolled in the same classes as the Uber-students. The "Class Peers" group is re-weighted by coarsened exact matching on observables to be more similar to the sample of Uber-students. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

Table G.5: Weekly Uber-student driving activity

	Mean	Std. Dev.
Activity and trips		
Hours online	18.25	(17.29)
Completed trips	36.62	(35.74)
Uber earnings		
Weekly earnings	442.21	(461.71)
Earnings per trip	12.37	(5.66)
Earnings per hour	24.26	(9.89)
Tips per trip	1.12	(1.10)
Tips per hour	2.20	(2.25)
Ratings		
Average weekly rating	4.94	(0.13)
Observations	28896	

Notes: Sample is limited to weeks in which drivers are enrolled in ASU classes. The variables "hours online", "completed trips", and "weekly earnings" include zeros for weeks that the drivers were inactive on the Uber platform.

Table G.6: Correlations of Uber activity measures for enrolled Uber-students

	Total pay (100s)	Incentive pay (100s)	Tips (100s)	Completed trips	Hours online
Total pay (100s)	1.000				
Incentive pay (100s)	0.674***	1.000			
Tips (100s)	0.598***	0.350***	1.000		
Completed trips	0.915***	0.596***	0.613***	1.000	
Hours online	0.880***	0.451***	0.585***	0.878***	1.000
Observations			27,475		

Notes: Table reports correlation coefficients across each metric of weekly Uber activity. "Total pay" corresponds to the driver's total weekly earnings from Uber, "Incentive pay" refers to pay received due to specific incentive programs (e.g., completing a target number of trips), "Tips" reflect the weekly earnings received in formal passenger tips, and "Completed trips" and "Hours online" denote the total number of trips completed and hours spent on the Uber application, respectively. Sample is limited to the observations included in the analytic sample used in our main regression specifications presented in Table 5. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

Table G.7: Sample comparison

	Uber-Students		**Class Peers**	
	Survey	Admin	Survey	Admin
Demographics				
Age	38.78	39.73	30.64	28.51
Female	0.37	0.15	0.62	0.61
White	0.47	0.36	0.58	0.58
Black or Hispanic	0.37	0.47	0.27	0.28
Academic Profile				
STEM degree	0.42	0.39	0.55	0.30
Observations	243	1540	344	64248

Notes: "Uber-students" denotes ASU students who participate in the ASU-Uber partnership. "Class peers" corresponds to all other students that are enrolled in the same classes as the Uber-students. "STEM degree" indicates if the student is in a science, technology, engineering, or math degree program.

Table G.8: Determinants of study effort (hours), week-year fixed effects included

	Study Hours in c
Avg. study hours in c, peers	1.119^{***}
	(0.027)
Sum of avg. study hours in $-c$, peers	-0.007
	(0.006)
Market level active hourly pay, currently elig.	-0.002
	(0.004)
Observations	55287
Individual Fixed Effects	✓
Classroom Fixed Effects	✓
Week-Year Fixed Effects	✓
Weather controls	✓
Mean dep. var.	1.58
Avg. study hours in c, peers	1.28
Sum avg. study hours in $-c$, peers	1.55
Market level active hourly pay, peers	33.56

Notes: The dependent variable is weekly study hours in course c. "Avg. study hours in c, peers" is a proxy for coursework that denotes the average Canvas hours of non-Uber-students in course c. "Sum of avg. study hours in $-c$, peers" denotes the sum of the average hours per week that peers spent on Canvas across all other courses besides c. "Market level active hourly pay, currently elig." denotes the average active hourly pay in a given week of drivers who are currently eligible for the ASU–Uber partnership, but not enrolled. Specifications include individual, classroom (e.g., a specific section of ECN 101), and week-year fixed effects. Weather controls include second-order polynomials in total weekly rainfall and snowfall. Each regression model also includes a flexible control for the number of days in the week for the course. Standard errors are clustered at the classroom level. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

Table G.9: Determinants of study effort (clicks and logins)

	(1) Weekly clicks	(2) Weekly logins	(3) Weekly clicks	(4) Weekly logins
Avg. study hours in c, peers	36.868***	2.105***	-	-
	(5.580)	(0.596)		
Sum of avg. study hours in $-c$, peers	-0.226	-0.048	-	-
	(0.374)	(0.038)		
Assignments graded in c	-	-	2.036***	0.220***
			(0.464)	(0.048)
Sum of assignments graded in $-c$	-	-	-0.531***	-0.036**
			(0.155)	(0.015)
Market level active hourly pay, currently elig.	-0.904***	-0.026	-1.616***	-0.068***
	(0.196)	(0.019)	(0.204)	(0.017)
Observations	55293	55293	55293	55293
First-stage F-statistic	-	-	586.82	586.82
Individual Fixed Effects	✓	✓	✓	✓
Classroom Fixed Effects	✓	✓	✓	✓
Month-Year Fixed Effects	✓	✓	✓	✓
Weather controls	✓	✓	✓	✓
Mean dep. var.	76.79	9.01	76.79	9.01
Avg. study hours in c, peers	1.28	1.28	-	-
Sum avg. study hours in $-c$, peers	1.55	1.55	-	-
Assignments graded in c	-	-	2.21	2.21
Sum assignments graded in $-c$	-	-	2.78	2.78
Market level active hourly pay, peers	33.56	33.56	33.56	33.56
Elasticity w.r.t. peer coursework	0.615	0.299	0.059	0.054
Elasticity w.r.t. peer earnings	-0.395	-0.098	-0.706	-0.255

Notes: The dependent variable in each column is the number of weekly clicks or logins by a given student on course c's Canvas page, as indicated by the column heading. "Avg. study hours in c, peers" is a proxy for coursework that denotes the average Canvas hours of non-Uber-students in course c. The variable "Sum of avg. study hours in $-c$, peers" denotes the sum of the average hours per week that peers spent on Canvas across all other courses besides c. "Assignments graded in c" and "Sum of assignments graded in $-c$" denote the number of assignments graded in course c and all other courses $-c$ for a student in a given week. The results in Columns (3) and (4) are from a 2SLS specification where "Assignments graded in c" and " Sum of assignments graded in $-c$" are instrumented with the average assignments graded by non-Uber peers in c and $-c$, respectively. "Market level active hourly pay, currently elig." denotes the average active hourly pay in a given week of drivers who are currently eligible for the ASU-Uber partnership, but not enrolled. Specifications include multiple sets of fixed effects. Weather controls include second-order polynomials in total weekly rainfall and snowfall. Each regression model also includes a flexible control for the number of days in the week for the course. Elasticities are calculated by multiplying the relevant coefficient estimate by the ratio of mean coursework or market-level hourly earnings to the mean of the dependent variable. Standard errors are clustered at the classroom level. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

Table G.10: Determinants of work effort (hours), lag and lead specifications

	(1) Lag specification	(2) Lead specification
Market-level hourly pay		
Current week	0.228***	0.121**
	(0.053)	(0.048)
Lag / lead week	-0.101**	-0.062
	(0.046)	(0.051)
Peer study hours across all courses		
Current week	-0.754***	-0.612***
	(0.093)	(0.076)
Lag / lead week	-0.054	0.022
	(0.063)	(0.061)
Obs.	24150	24496
Individual fixed effects	Yes	Yes
Course bundle fixed effects	Yes	Yes
Month-year fixed effects	Yes	Yes
City fixed effects	Yes	Yes
Weather controls	Yes	Yes

Note: The dependent variable is weekly Uber driving hours. Standard errors clustered at the individual level are reported in parentheses. "Market-level hourly pay" denotes the average active hourly pay in a given week among drivers eligible for the ASU–Uber partnership but not enrolled. "Peer study hours across all courses" denotes the sum, across all courses, of peers' average weekly Canvas hours. Specifications include multiple sets of fixed effects; course-bundle fixed effects correspond to the set of courses taken in a given term. Weather controls include second-order polynomials in total weekly rainfall and snowfall. The lag and lead specification columns report point estimates from the following regression models:

$$WorkE_{i,t} = \beta_0 + \beta_1 \sum_{c \in C_{i,t}} CourseWork_{c,t} + \beta_2 \sum_{c \in C_{i,t+s}} CourseWork_{c,t+s} + \beta_3 MarketHourlyPay_{l(i,t),t}$$
$$+ \beta_4 MarketHourlyPay_{l(i,t),t+s} + \beta_5 Weather_{l(i,t),t} + \psi_i + \delta_m + \phi_b + \gamma_{l(i,t)} + \epsilon_{i,t},$$

where $s = -1$ for the lag specification and $s = 1$ for the lead specification. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

Table G.11: Determinants of study effort (hours), lag and lead specifications

	(1) Lag specification	(2) Lead specification
Market-level hourly pay		
Current week	-0.002	-0.010***
	(0.004)	(0.004)
Lag / lead week	-0.006*	-0.001
	(0.004)	(0.004)
Peer Canvas hours in same course		
Current week	1.121***	1.165***
	(0.028)	(0.026)
Lag / lead week	0.054***	0.002
	(0.015)	(0.022)
Peer Canvas hours in other courses		
Current week	0.003	-0.005
	(0.008)	(0.008)
Lag / lead week	-0.014*	-0.010
	(0.008)	(0.016)
Obs.	47954	48799
Individual fixed effects	Yes	Yes
Classroom fixed effects	Yes	Yes
Month-year fixed effects	Yes	Yes
Weather controls	Yes	Yes
Week-length controls	Yes	Yes

Notes: The dependent variable is weekly Canvas hours. Standard errors clustered at the classroom level are reported in parentheses. "Market-level hourly pay" denotes the average active hourly pay in a given week among drivers eligible for the ASU–Uber partnership but not enrolled. "Peer Canvas hours in same course" is a proxy for coursework that denotes the average Canvas hours of non-Uber-students in the course. The variable "Peer Canvas hours in other courses" denotes the sum of the average hours per week that peers spent on Canvas across all other courses the student is enrolled in at the same time. Specifications include multiple sets of fixed effects. Weather controls include second-order polynomials in total weekly rainfall and snowfall. Each regression model also includes a flexible control for the number of days in the week for the course. The lag and lead specification columns report point estimates from the following regression models:

$$CourseE_{i,c,t} = \delta_0 + \delta_1 CourseWork_{i,t,c} + \delta_2 CourseWork_{i,t+s,c} + \delta_3 \sum_{-c} CourseWork_{i,t,-c}$$
$$+ \delta_4 \sum_{-c} CourseWork_{i,t+s,-c} + \delta_5 MarketHourlyPay_{l(i,t),t} + \delta_6 MarketHourlyPay_{l(i,t),t+s}$$
$$+ \delta_7 Weather_{l(i,t),t} + \delta_8 X_{c,t} + \pi_i + \alpha_m + \rho_c + \eta_{i,c,t}$$

where $s = -1$ for the lag specification and $s = 1$ for the lead specification. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

Table G.12: Heterogeneity in the determinants of work and study hours

	(1)	(2)	(3)	(4)	(5)	(6)
	Full-time driver	STEM	Female	High-need	High Incoming GPA	Age $\geq$ 35
Panel A: Work hours						
Market level active hourly pay, currently elig.	0.124**	0.138**	0.177***	0.134	0.097	0.143*
	(0.057)	(0.063)	(0.057)	(0.083)	(0.071)	(0.081)
Market level active hourly pay, currently elig. × var. in top row	0.096	0.090	-0.052	0.046	0.091	0.025
	(0.112)	(0.115)	(0.109)	(0.102)	(0.106)	(0.100)
Sum avg. study hours all c, peers	-0.381***	-0.360***	-0.497***	-0.554***	-0.465***	-0.489***
	(0.090)	(0.092)	(0.094)	(0.132)	(0.115)	(0.118)
Sum avg. study hours all c, peers × var. in top row	-0.237*	-0.285**	0.166	0.079	0.027	-0.005
	(0.137)	(0.136)	(0.157)	(0.144)	(0.134)	(0.132)
Observations	27404	26636	26690	27388	23902	27404
Panel B: Study hours						
Market level active hourly pay, currently elig.	-0.010**	-0.007	-0.007*	-0.011*	-0.007	-0.010*
	(0.004)	(0.005)	(0.004)	(0.006)	(0.005)	(0.005)
Market level active hourly pay, currently elig. × var. in top row	0.006	-0.001	-0.015	0.003	-0.003	0.004
	(0.006)	(0.006)	(0.010)	(0.007)	(0.006)	(0.006)
Avg. study hours in c, peers	1.041***	1.074***	1.077***	0.978***	1.023***	1.000***
	(0.076)	(0.042)	(0.061)	(0.166)	(0.063)	(0.048)
Avg. study hours in c, peers × var. in top row	0.207*	0.156**	0.315	0.205	0.107*	0.225***
	(0.124)	(0.063)	(0.243)	(0.198)	(0.055)	(0.047)
Sum of avg. study hours in $-c$, peers	-0.006	0.002	-0.003	0.004	-0.019**	-0.009
	(0.008)	(0.008)	(0.008)	(0.018)	(0.008)	(0.008)
Sum of avg. study hours in $-c$, peers × var. in top row	-0.008	-0.027**	-0.045	-0.017	0.009	0.001
	(0.013)	(0.013)	(0.033)	(0.022)	(0.012)	(0.011)
Observations	55293	53731	53975	55266	48263	55293

Notes: "Full-time drivers" are those that average at least 25 hours of driving per week when not enrolled in ASU classes. "High-need" indicates that Uber-students are either "High" or "Very High" need based on their FAFSA records, and "High incoming GPA" indicates that the student's incoming GPA was at least 2.72 (the sample median). For Panel A—the dependent variable in each column is hours working on the Uber platform. "Market level active hourly pay, currently elig." denotes the average active hourly pay in a given week of drivers who are currently eligible for the ASU-Uber partnership but not enrolled. "Sum of avg. study hours across all c, peers" denotes the sum of the average hours per week that peers spent on Canvas across all courses. For Panel B—the dependent variable in each column is weekly study hours in course c. "Avg. study hours in c, peers" is a proxy for coursework that denotes the average Canvas hours of non-Uber-students in course c. The variable "Sum of avg. study hours in $-c$, peers" denotes the sum of the average hours per week that peers spent on Canvas across all other courses besides c. All specifications include multiple sets of fixed effects, second-order polynomials in total weekly rainfall and snowfall. Each regression model in Panel B also includes a flexible control for the number of days in the week for the course. In Panel A, standard errors are clustered at the individual level. In Panel B, standard errors are clustered at the classroom level.*, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

Table G.13: The effect of work and study effort on completion rates

	(1) Completed Course	(2) Completed Course
Total work hours	0.001	
	(0.001)	
Total study hours		0.039***
		(0.005)
Obs.	4259	4259
First-stage F-statistic	23.80	73.61
Classroom Fixed Effects	✓	✓
Student-by-Term Fixed Effects	✓	✓
Avg. dep. var.	0.81	
Avg. total hours on Canvas	13.50	
Avg. total Uber hours	142.16	

Notes: The variable "Total work hours" denotes the total hours an Uber-student spent connected to the Uber platform throughout the course. "Total study hours" refers to the sum of individual Uber-student's own Canvas hours throughout a given course. Each regression model includes classroom (e.g., a specific section of ECN 101), and individual-by-term fixed-effects. The sample excludes all terms that were meaningfully disrupted by the COVID-19 pandemic. The model in Column (1) is estimated via 2SLS where "Total work hours" is instrumented with the average (across weeks) of average market-level pay for Uber drivers who were eligible for the ASU-Uber program but did not enroll. The model in Column (2) is estimated via 2SLS where "Total Canvas hours" is instrumented with the sum (across weeks) of average Canvas hours for non-Uber peers in the course. All standard errors are clustered at the classroom level. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

Table G.14: The effect of work and study effort on course pass rate

	(1) Passed Course	(2) Passed Course
Total work hours	-0.000	
	(0.001)	
Total study hours		0.032***
		(0.005)
Obs.	4259	4259
First-stage F-statistic	23.80	73.61
Classroom Fixed Effects	✓	✓
Student-by-Term Fixed Effects	✓	✓
Avg. dep. var.	0.75	
Avg. total hours on Canvas	13.50	
Avg. total Uber hours	142.16	

Notes: The variable "Total work hours" denotes the total hours an Uber-student spent connected to the Uber platform throughout the course. "Total study hours" refers to the sum of individual Uber-student's own Canvas hours throughout a given course. Each regression model includes classroom (e.g., a specific section of ECN 101), and individual-by-term fixed-effects. The sample excludes all terms that were meaningfully disrupted by the COVID-19 pandemic. The model in Column (1) is estimated via 2SLS where "Total work hours" is instrumented with the average (across weeks) of average market-level pay for Uber drivers who were eligible for the ASU-Uber program but did not enroll. The model in Column (2) is estimated via 2SLS where "Total Canvas hours" is instrumented with the sum (across weeks) of average Canvas hours for non-Uber peers in the course. All standard errors are clustered at the classroom level. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

Table G.15: Expected Graduation Probabilities and Earnings in 6 Years

	Uber-Students	Class Peers	P-values of the Difference Uber-Students - Class Peers
Prob. complete studies in 6 years (%)	91.70	92.90	0.41
Prob. complete studies in 6 years (%) if remained at prev. college†	62.07	70.41	0.05
Expected returns from finishing the degree (ratio)	1.99	1.85	0.16
Observations	243	344	

Notes: "Uber-students" denotes ASU students who participate in the ASU-Uber partnership. "Class peers" corresponds to all other students that are enrolled in the same classes as the Uber-students. "Expected returns from finishing the degree" refers to the ratio of "Expected annual income in 6 years if graduate" over "Expected annual income in 6 years if dropout today". † These probabilities are calculated conditional on students who transferred institutions. P-values denote whether differences between Uber-students and class peers are statistically significant.

Table G.16: Higher Education Quality Comparison - Transfer Students

	Uber-students	Class Peers	Diff: P-values
ASU Better Educ.	0.48	0.57	0.08
ASU Inferior Educ.	0.06	0.08	0.62
ASU Similar Educ.	0.46	0.35	0.04
Observations	125	256	

Notes: "Uber-students" denotes ASU students that participate in the ASU-Uber partnership. "Class peers" corresponds to students that are enrolled in the same classes as the Uber students. The survey question and choices were *"How would you compare the quality of education at ASU to that of your previous institution? ASU offers a better/inferior/similar education than my previous institution."*

Table G.17: ASU Credits Earned and Graduation Rates by Cohort

Academic Year Cohort	Cohort Size	Mean ASU Only Credits Earned	Graduation Rate to Date	5-Year Graduation Rate
2018–2019	147	78.79	26.5%	23.1%
2019–2020	1079	80.38	30.7%	28.5%
2020–2021	568	83.10	23.6%	-
2021–2022	386	80.61	22.9%	-
Cohort-Size Weighted	2180	81.02	27.2%	27.9%*

Notes: "Graduation rate to date" refers to the share of students observed to have graduated as of December 2025. "5-year graduation rate" refers to the share of students who graduated within five years of initial cohort entry. "Mean ASU Only Credits Earned" excludes transfer credits. The final row reports cohort-size-weighted averages across cohorts.

* Cohort-size-weighted 5-year graduation rate excludes members of the 2020-21 and 2021-22 cohorts since many of those students had not been enrolled for at least five years by December 2025.

Table G.18: Determinants of work effort, excluding likely dropouts

	(1) Work Hours	(2) Work Hours	(3) Work Hours	(4) Completed Trips	(5) Completed Trips	(6) Completed Trips
Sum of avg. study hours across all c, peers	-0.686***	-0.683***	-0.677***	-1.286***	-1.280***	-1.337***
	(0.073)	(0.073)	(0.101)	(0.147)	(0.147)	(0.206)
Market level active hourly pay, currently elig.	0.108**	0.115**	–	0.301***	0.326***	–
	(0.051)	(0.052)		(0.113)	(0.113)	
Obs.	25972	25907	24777	25972	25907	24777
Individual Fixed Effects	✓	✓	✓	✓	✓	✓
Course Bundle Fixed Effects	✓	✓	✓	✓	✓	✓
Month-Year Fixed effects	✓	✓		✓	✓	
City Fixed Effects	✓	✓		✓	✓	
Weather controls		✓			✓	
City-Week FE			✓			✓
Mean dep. var.	18.01	17.98	18.19	36.09	36.03	36.38
Market level active hourly pay, peers	33.64	33.63		33.64	33.63	
Sum avg. study hours all c, peers	2.48	2.48	2.49	2.48	2.48	2.49
Mean hourly earnings	24.22	24.21	24.37	24.22	24.21	24.37
Mean pay-per-trip	12.37	12.37	12.45	12.37	12.37	12.45
Elasticity w.r.t. peer coursework	-0.094	-0.094	-0.093	-0.088	-0.088	-0.091
Elasticity w.r.t. peer earnings	0.201	0.216		0.280	0.304	

Notes: Columns (1) to (3) include as dependent variable weekly work hours, while columns (4) to (6) include completed trips. The variable "Sum of avg. study hours across all c, peers" denotes the sum of the average hours across courses that peers spent on Canvas per week. The variable "Market level active hourly pay, currently elig." denotes the average hourly pay of drivers (currently eligible) in a given week. Likely dropouts are excluded from the estimating sample. Additional details on how we identify likely dropouts are provided in Appendix A.2. Specifications include multiple sets of fixed effects. Course bundle fixed effects denote the set of courses taken in a given term. Weather controls include second-order polynomials in rainfall and snowfall. Standard errors are clustered at the individual level. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

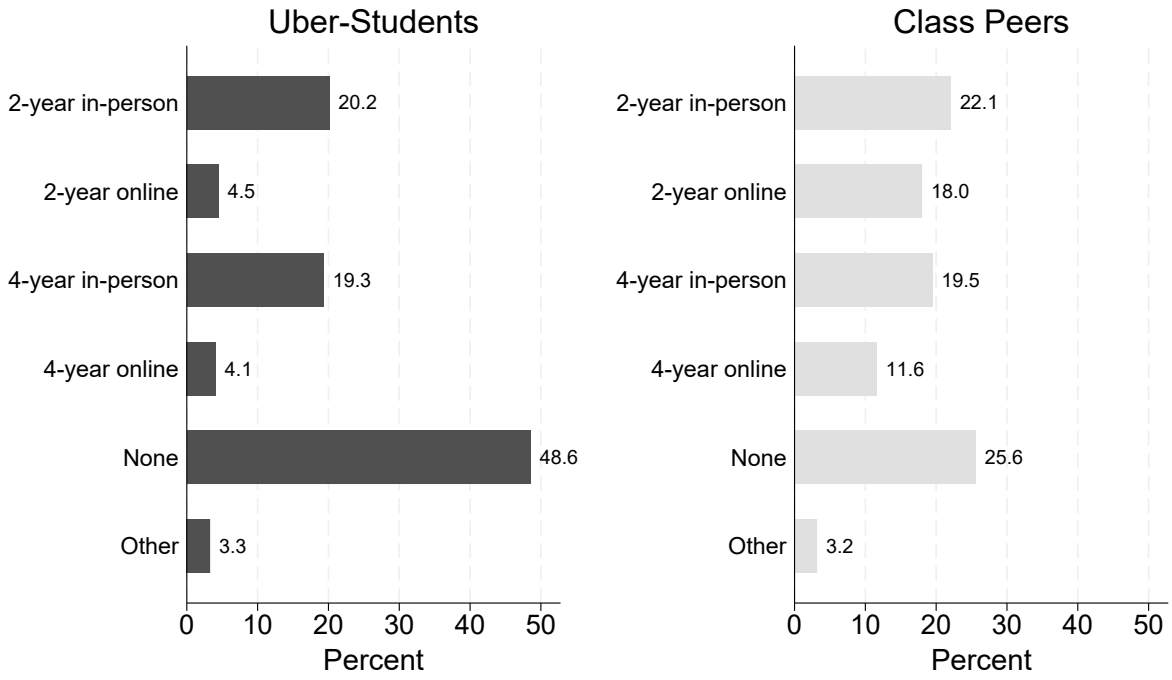
Table G.19: Determinants of study effort, excluding likely dropouts

	(1) Study Hours in c	(2) Study Hours in c
Avg. study hours in c, peers	1.206***	-
	(0.028)	
Sum of avg. study hours in $-c$, peers	-0.011*	-
	(0.006)	
Assignments graded in c	-	0.045***
		(0.015)
Sum of assignments graded in $-c$	-	-0.012**
		(0.005)
Market level active hourly pay, currently elig.	-0.005	-0.028***
	(0.003)	(0.004)
Observations	51385	51385
First-stage F-statistic	-	385.86
Individual Fixed Effects	✓	✓
Classroom Fixed Effects	✓	✓
Month-Year Fixed effects	✓	✓
Weather controls	✓	✓
Mean dep. var.	1.70	1.70
Avg. study hours in c, peers	1.28	-
Sum avg. study hours in $-c$, peers	1.57	-
Assignments graded in c	-	2.38
Sum assignments graded in $-c$	-	2.87
Market level active hourly pay, peers	33.57	33.57
Elasticity w.r.t. peer coursework	0.908	0.063
Elasticity w.r.t. peer earnings	-0.090	-0.556

Notes: The dependent variable in each column is weekly study hours in course c. "Avg. study hours in c, peers" is a proxy for coursework that denotes the average Canvas hours of non-Uber-students in course c. The variable "Sum of avg. study hours in $-c$, peers" denotes the sum of the average hours per week that peers spent on Canvas across all other courses besides c. "Assignments graded in c" and "Sum of assignments graded in $-c$" denote the number of assignments graded in course c and all other courses $-c$ for a student in a given week. The results in Column (2) are from a 2SLS specification where "Assignments graded in c" and " Sum of assignments graded in $-c$" are instrumented with the average graded assignments for non-Uber peers in c and $-c$, respectively. "Market level active hourly pay, currently elig." denotes the average active hourly pay in a given week of drivers who are currently eligible for the ASU-Uber partnership, but not enrolled. Likely dropouts are excluded from the estimating sample. Additional details on how we identify likely dropouts are provided in Appendix A.2. Specifications include multiple sets of fixed effects. Weather controls include second-order polynomials in total weekly rainfall and snowfall. Each regression model also includes a flexible control for the number of days in the week for the course. Standard errors are clustered at the classroom level. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

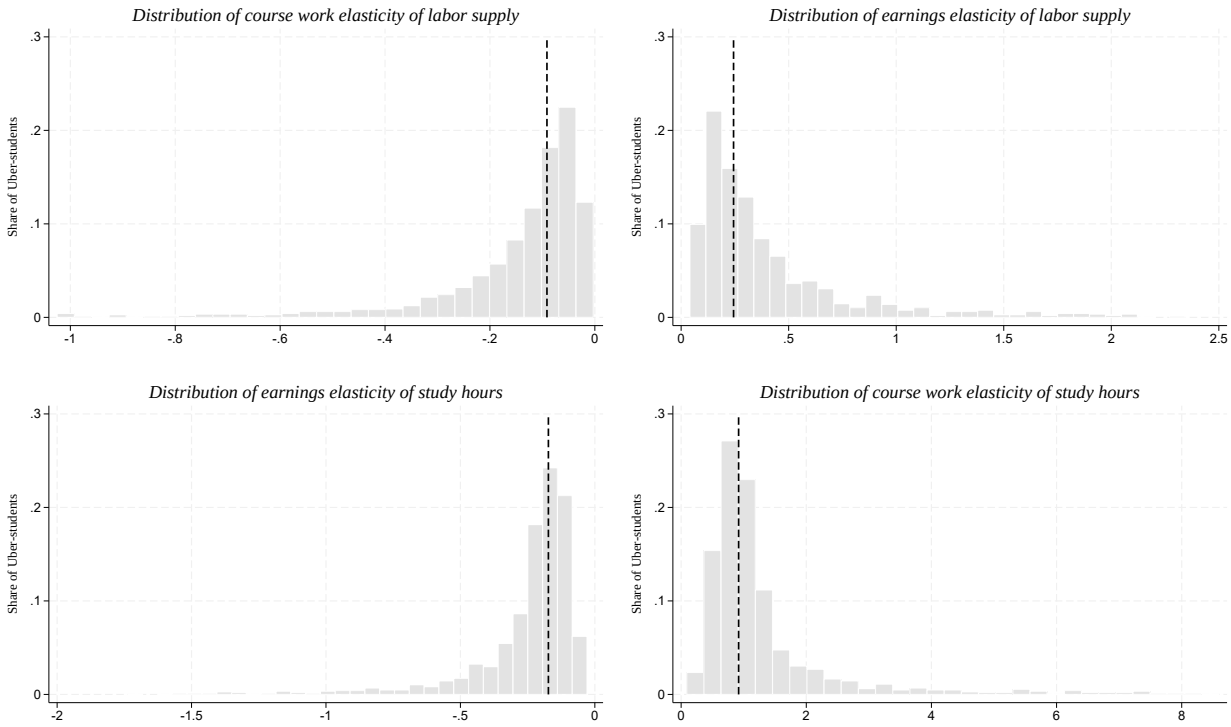
H Appendix Figures

Figure H.1: Previous Institution Attended



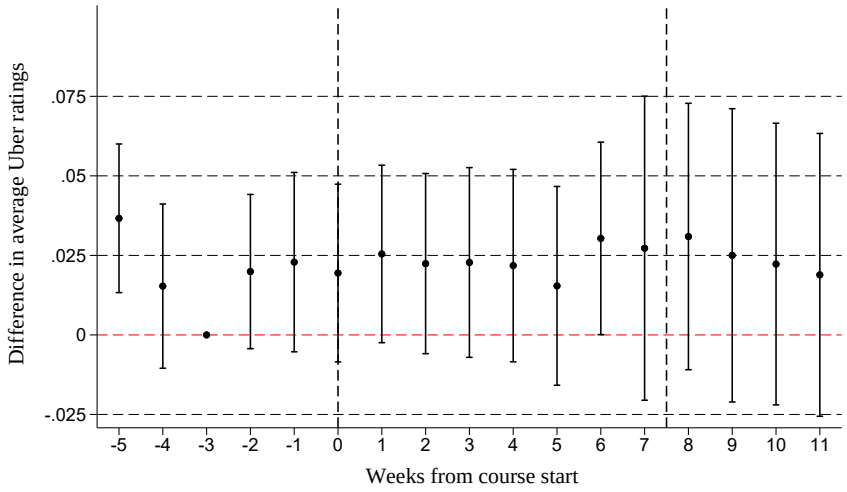
Notes: "Uber-students" denotes ASU students who participate in the ASU-Uber partnership. "Class peers" corresponds to all other students that are enrolled in the same classes as the Uber-students. There are 243 Uber-students and 344 Class peers. The corresponding survey question is: *"You indicated that you were enrolled at another institution. Which of the following best describes your previous enrollment? I attended a 4-year/2-year public/private institution and took in-person/online classes."* We grouped the options of public and private institutions together for analysis.

Figure H.2: Distribution of Elasticities



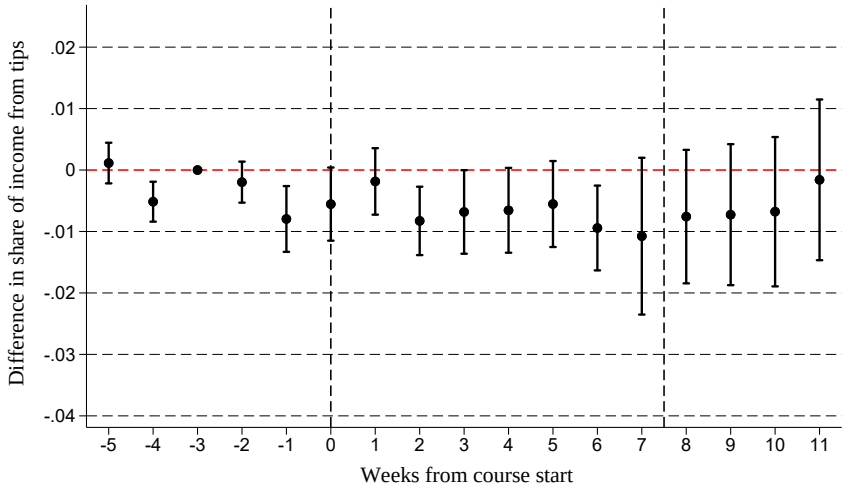
Notes: Distribution of elasticities using individual-level means of relevant variables (i.e., average market-level pay, peer course hours, own driving, and study hours while enrolled at ASU).

Figure H.3: Variation in driver ratings during eight-week courses



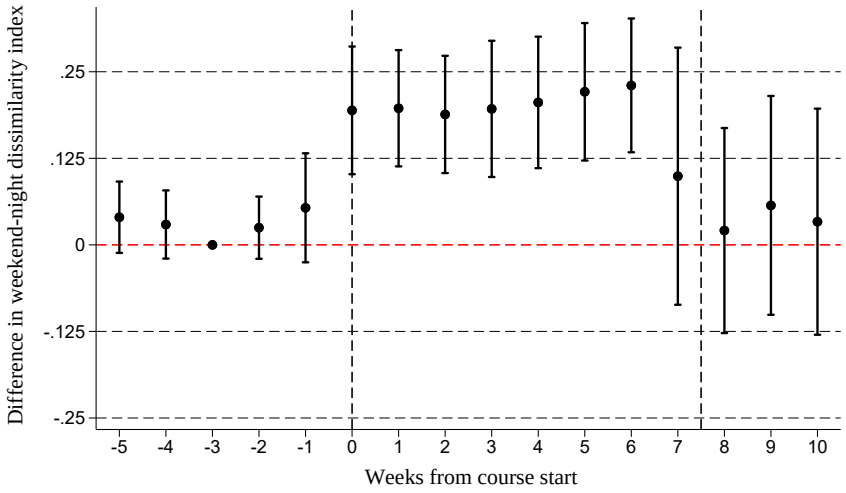
Notes: OLS coefficient estimates from a version of Equation (1) where the dependent variable is average driver rating and 95% confidence intervals are plotted. The estimating sample excludes individuals who appear to have dropped out of the course. Standard errors are clustered at the classroom level.

Figure H.4: Variation in the share of pay coming from tips during eight-week courses



Notes: OLS coefficient estimates from a version of Equation (1) where the dependent variable is the share of weekly earnings coming from tips and 95% confidence intervals are plotted. The estimating sample excludes individuals who appear to have dropped out of the course. Standard errors are clustered at the classroom level.

Figure H.5: Variation in flexibility on weekend peak periods during eight-week courses



Notes: OLS coefficient estimates from a version of Equation (1) where the dependent variable is the dissimilarity index for share of driving hours allocated in Friday and Saturday afternoons and nights (i.e., 12:00PM-12:00AM) and 95% confidence intervals are plotted. The estimating sample excludes individuals who appear to have dropped out of the course. Standard errors are clustered at the classroom level.

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